

REVIEW ARTICLE

Lattice-Based Public Auditing Schemes for Cloud Storage Security: A Comprehensive Survey

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ABSTRACT

Public auditing is a method used to verify the integrity of data stored in the cloud without requiring access to the actual data. However, the advancement of quantum computers poses significant security threats to existing public auditing schemes, as these schemes are based on conventional cryptography hard problems, which are vulnerable to quantum attacks. To address this, NIST has launched the development of post-quantum cryptographic primitives and protocols. Among the various approaches, lattice-based cryptography (LBC) is considered one of the most promising candidates due to its strong security guarantees and inherent resistance to quantum attacks. Leveraging LBC, several researchers have proposed lattice-based public auditing (LBPA) schemes for cloud storage security based on lattice hardness assumptions. This paper provides a comprehensive survey of existing LBPA schemes for cloud storage, presenting a detailed taxonomy and analyzing their similarities, differences, and performance. Additionally, it highlights key challenges and outlines future research directions for designing efficient and secure public auditing schemes in the post-quantum era.

1 | Introduction

Cloud storage is a key component of modern cloud computing services, enabling individuals and organizations to outsource data to cloud servers instead of relying on local systems. This outsourcing offers several significant advantages, including cost reduction, accessibility, scalability, flexibility, and minimal maintenance overhead. With these benefits, cloud storage adoption has increased rapidly across various sectors. Prominent cloud storage providers (CSP) such as Google Drive, iCloud, and Microsoft OneDrive serve millions of users worldwide [1]. Despite its benefits, cloud storage raises critical security issues related to data confidentiality, integrity, and availability [2].

Among these, data integrity is a primary concern due to the following factors: Once data is outsourced, the user relinquishes direct control over it, raising concerns about its correctness and intactness. The data is susceptible to accidental or intentional modification, corruption, or deletion, which may arise from poor system maintenance or cost-saving measures by the CSP [3]. In certain cases, the CSP might delete infrequently accessed data or conceal data loss incidents to preserve its reputation and avoid service-level penalties. As a result, the data owner (DO) could download and verify all data periodically. However, this approach is inefficient and leads to high computational cost. To address this, researchers have proposed the “Remote Data Integrity Checking (RDIC) method,” which permits users to

check the integrity of outsourced data without retrieving the entire file [4, 5]. RDIC schemes are broadly classified into private/public auditing.

In private auditing, the DO checks the integrity of their data stored in the cloud. In this process, first, the DO generates signatures for the data blocks and subsequently uploads both the data and signatures to the cloud. Thereafter, the DO verifies the data integrity through a challenge–response protocol, in which the DO sends a challenge message to the CSP. In response, the CSP generates and returns an integrity proof based on the challenge. However, the DO may not be able to perform integrity checks regularly due to the potentially large volume of data files, a busy schedule, or limited knowledge of the verification procedures. As a result, the integrity checking process may become inefficient or yield biased outcomes. To address these limitations, public auditing (PA) has been introduced.

In PA, on behalf of the DO, a trusted Third-Party Auditor (TPA) verifies data integrity. Upon obtaining a request to audit, the TPA sends a challenge to the CSP and analyzes the proof provided. This approach reduces the computational burden on the DO and ensures continuous and reliable auditing. The TPA is assumed to be professional, independent, and well-equipped to handle the verification process [6].

Over the last decade, a wide range of public auditing schemes [7–12] have been proposed based on traditional cryptographic assumptions, such as the Integer Factorization (IF), Discrete Logarithm Problem (DLP), and Diffie-Hellman Problem (DHP). However, these schemes face a critical threat in the emerging quantum computing era. In 1994, Peter Shor proposed a polynomial-time quantum algorithm for factoring large integers, which effectively breaks the security of RSA [13, 14]. Consequently, all PA schemes built on classical number-theoretic problems are considered quantum-insecure.

To mitigate this threat, the “National Institute of Standards and Technology (NIST) launched a Post-Quantum Cryptography (PQC) standardization project. It has identified five primary categories of PQC candidates: lattice-based cryptography (LBC), code-based cryptography, multivariate polynomial cryptography, hash-based cryptography, and isogeny-based cryptography.” Among these, LBC stands out as the most promising due to its worst-case hardness guarantees, asymptotic efficiency, and versatility across a wide range of cryptographic primitives. The first lattice-based construction was proposed by Milkós Ajtai in 1996 [15], with security reductions to the “Small Integer Solution (SIS) and Inhomogeneous Small Integer Solution (ISIS) problems.” Later, in 2005, Oded Regev introduced the Learning With Errors (LWE) problem and its variant Ring-LWE, further expanding the foundations of LBC [16]. Based on these (SIS (ISIS) and R-LWE) lattice hardness assumptions, numerous lattice-based public auditing (LBPA) schemes have been proposed [17–44].

1.1 | Main Contributions

- In this survey, we design a taxonomy of LBPA schemes, classifying them into four categories: Public Key Infrastructure (PKI)-based, Identity-Based Cryptography (IBC)-based,

Certificateless Cryptography (CLC)-based, and Attribute-Based Cryptography (ABC)-based techniques. It also describes the system architecture, key algorithms, and description of each technique.

- This survey also provides a comparative analysis of existing LBPA schemes with respect to security properties and performance metrics, offering valuable insights for the design of more secure and efficient solutions.
- This survey highlights challenges in existing LBPA schemes such as resistance to malicious auditors and CSPs and trapdoor generation efficiency. It also outlines promising directions for future research into efficient and quantum-resistant public auditing mechanisms.
- To the extent of our knowledge, this is the first comprehensive and dedicated survey focused on LBPA schemes for cloud storage security, addressing both theoretical foundations and practical considerations.

1.2 | Organization of Paper

The rest of the paper is organized as follows: Section 2 introduces the mathematical background. Section III presents the taxonomy. A comparative analysis in Section 4. Section 5 discusses the challenges and future directions. We conclude this survey with Section 6.

2 | Mathematical Background

2.1 | Lattice

Lattices have evolved from classical mathematical structures introduced by Minkowski in the early 20th century [45] to the backbone of modern PQC. Their cryptographic importance emerged with Ajtai’s seminal worst-case to average-case reduction [15], demonstrating that average-case instances of lattice problems inherit the quantum-resistant hardness of SVP and CVP. This breakthrough was strengthened by Micciancio and Goldwasser’s formalization of lattice algorithms and complexity theory [46], and Regev’s introduction of the LWE problem along with a proof of its quantum hardness [16].

Over the past decade, research has advanced rapidly through efficient structured variants such as R-LWE [47], Module-LWE [48], and worst-case Gaussian reduction techniques [49]. These developments, combined with trapdoor sampling techniques like GPV [50] and improved lattice trapdoors by Micciancio and Peikert [51], have enabled practical encryption, signatures, identity-based systems, and certificateless PQC. Together, these contributions form the modern landscape of lattice-based cryptography, characterized by provable security, asymptotic efficiency, and robustness against quantum algorithms.

A lattice is defined as a discrete additive subgroup of $\mathbb{R}^n$ generated by integer combinations of linearly independent basis vectors.

Formally, let $C = [c_1, c_2, \dots, c_m] \in \mathbb{R}^{m \times m}$ be a full-rank matrix. The lattice generated by C , denoted $L(C)$, is defined as:

$$L(C) = \left\{ xC = \sum_{i=1}^m x_i c_i : x_i \in \mathbb{Z} \right\}$$

where $x = (x_1, x_2, \dots, x_m)^\top \in \mathbb{Z}^m$ is an integer vector [50].

The hardness of fundamental lattice problems—SIS, ISIS, LWE, R-LWE, and M-LWE—underpins a wide range of cryptographic primitives, including collision-resistant hashing [52], signatures, homomorphic verifiable computations, and post-quantum encryption. Their algebraic structure enables fast polynomial-time operations, error-tolerant aggregation, and efficient trapdoor sampling [53].

Overall, the evolution from Minkowski's geometric foundations to modern hardness assumptions such as SIS, LWE, R-LWE, and M-LWE demonstrates the maturity of lattice theory. These advances have turned lattices into a reliable and versatile basis for PQC, offering strong worst-case security, efficient algebraic operations, and broad applicability across modern cryptographic constructions.

In essence, the progression from classical lattice geometry to modern SIS/LWE-based assumptions highlights the strength, efficiency, and quantum resistance of LBC, making it a dependable foundation for next-generation secure systems.

2.2 | LBC Hard Problems

- **SIS:** Given an integer modulus p , a positive real bound ρ , and a uniformly random matrix $B \in \mathbb{Z}_p^{n \times m}$, the goal is to find a nonzero integer vector $u \in \mathbb{Z}^m$ such that:

$$Bu = 0 \pmod{p} \quad \text{and} \quad \|u\| \leq \rho$$

where $\|u\|$ denotes the Euclidean (ℓ_2) norm of the vector u [50].

- **ISIS:** Given a prime p , a matrix $B \in \mathbb{Z}_p^{n \times m}$, a vector $v \in \mathbb{Z}_p^n$, and a positive real constant ρ , the goal is to find an integer vector $u \in \mathbb{Z}^m$ such that:

$$Bu = v \pmod{p} \quad \text{and} \quad \|u\| \leq \rho$$

where v is a known vector and u is the solution vector [50].

- **R-LWE:** Let y be a polynomial of any size in H_p (the polynomial ring $H_p = \mathbb{Z}_p[x]/\langle f \rangle$).

- **Search variant:** It is difficult to find y from arbitrary pairs $(b_i, b_i y + u_i)$, where $b_i \leftarrow H_p$ and $u_i \leftarrow D_{H_p, \delta}$, with a relatively small δ [37].
- **Decision variant:** It is hard to distinguish random pairs $(b_i y + u_i)$ from a_i , where $a_i \leftarrow H_p$.

2.3 | Trapdoor Generation Algorithms

- **TrapGen:** The TrapGen algorithm takes inputs (p, n) [15, 50] and outputs a matrix $B \in \mathbb{Z}_p^{n \times m}$ and a matrix $T_B \in \mathbb{Z}_p^{m \times m}$. Here, B is statistically close to $\mathbb{Z}_p^{n \times m}$, and T_B is a short lattice basis of the lattice $\Lambda_p^\perp(B)$ [26].
- **SamplePre:** Given a prime p and an integer $m \geq [2n \log p]$, the algorithm SamplePre(B, T_B, y, α) [15, 29] takes as inputs a matrix $B \in \mathbb{Z}_p^{n \times m}$, a short lattice basis $T_B \in \mathbb{Z}_p^{m \times m}$, a vector $y \in \mathbb{Z}_p^n$, and a parameter $\alpha \geq \|T_B\| \omega(\sqrt{\log m})$. It produces a sample from a distribution that is statistically close to the discrete Gaussian distribution $D_{\Lambda_p^\perp(B), \alpha}$, defined over the lattice $\Lambda_p^\perp(B)$ with parameter α .
- **NewBasisDel:** The NewBasisDel algorithm [25] takes as inputs a matrix $B \in \mathbb{Z}_p^{n \times m}$, a short lattice basis $T_B \in \mathbb{Z}_p^{m \times m}$ of the lattice $\Lambda_p^\perp(B)$, and an invertible matrix $R \in \mathbb{Z}_p^{m \times m}$ sampled from the distribution $D_{\mathbb{Z}^m, \delta_R}$ defined as $(D_{\mathbb{Z}^m, \delta_R})^m$, along with a Gaussian parameter $\delta \geq \|T_B\| \delta_R \sqrt{m\omega} \left(\sqrt{\log^3 m} \right)$. The algorithm outputs a randomized lattice basis T of $\Lambda_p^\perp(BR^{-1})$.

3 | Taxonomy and Evaluation of LBPA Schemes

3.1 | Taxonomy

Figure 1 shows the taxonomy of LBPA schemes. In this taxonomy, we classify LBPA schemes based on cryptographic techniques such as PKI-based, IBC-based, CLC-based, and ABC-based approaches.

3.1.1 | Evolution Timeline of LBPA Schemes (2014–2024)

Figure 2 illustrates the chronological evolution of LBPA schemes from 2014 to 2024 across different cryptographic

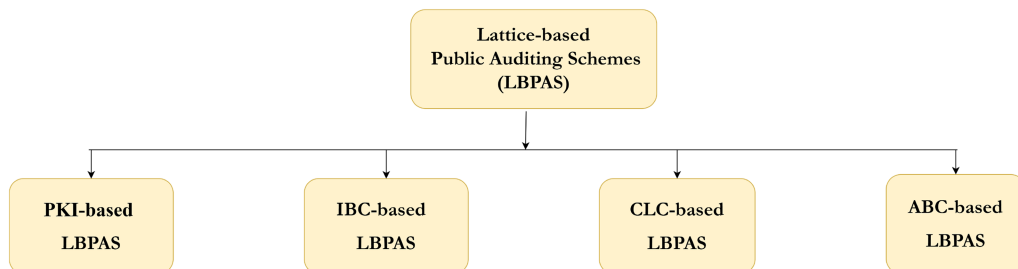


FIGURE 1 | Taxonomy of existing LBPA schemes.

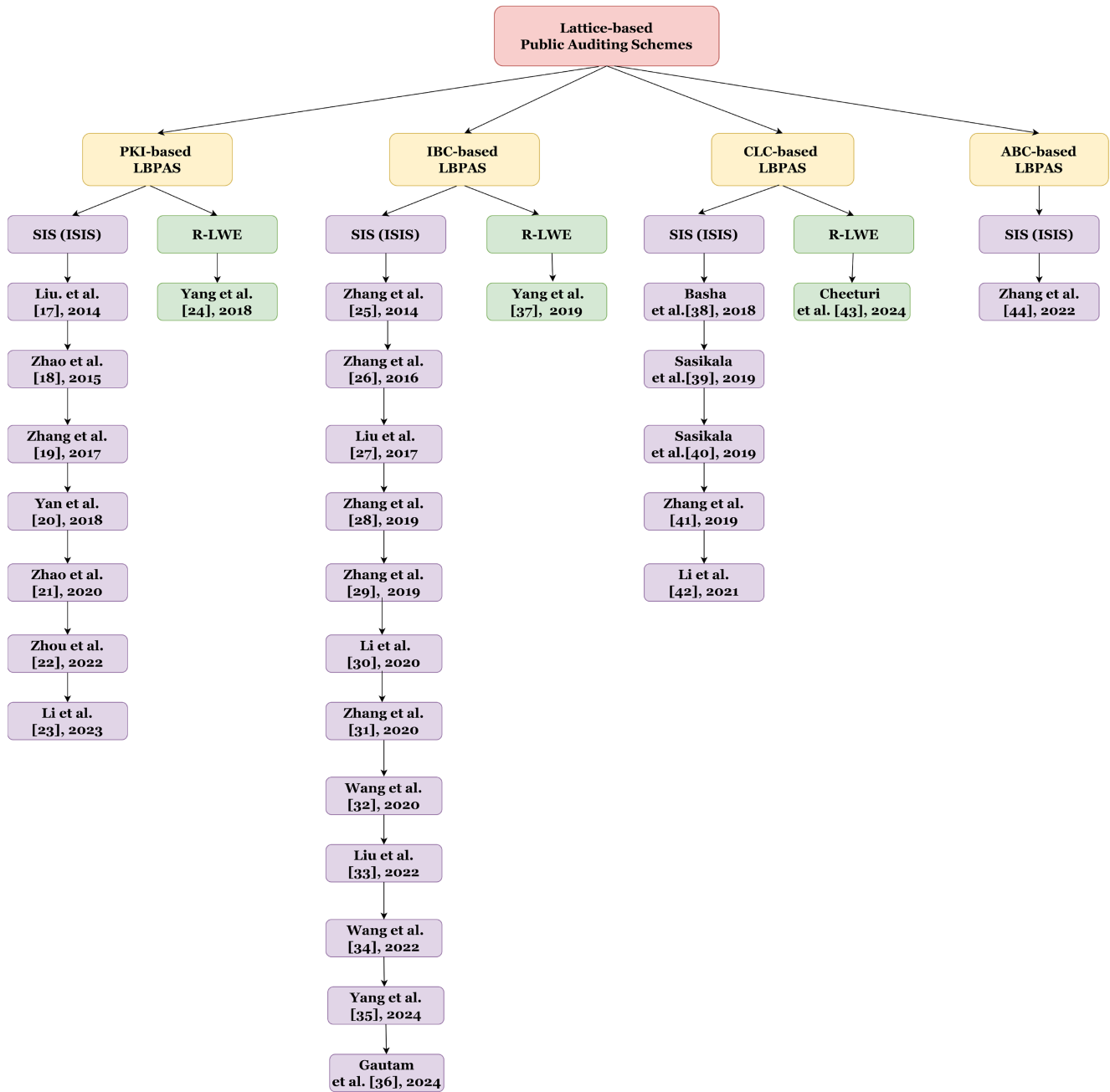


FIGURE 2 | Chronological evolution of LBPA schemes (2014–2024).

settings—PKI-based, IBC-based, CLC-based, and ABC-based models. It highlights the gradual transition from early SIS/ISIS-based constructions, focused on static integrity verification, to more advanced R-LWE-based schemes emphasizing efficiency, dynamic auditing, and quantum resistance. The timeline clearly depicts how LBPA research has matured toward stronger security assumptions and diversified architectures, forming the foundation for next-generation post-quantum auditing mechanisms.

- *PKI-based method*: PKI is one of the most commonly used technologies, extensively utilized across various industries. A certificate authority (CA) is the trusted entity in a PKI-based method that authenticates documents and binds public keys to DOs. In PKI systems, a DO must first

create a public/secret key pair (pk , sk), then send the pk to the CA and request a pk certificate while keeping the sk confidential.

- *IBC-based method*: IBC was coined by Shamir in 1985 [38]. In an IBC method, the DO's identity, such as their name or identity card number, serves as their pk . This eliminates the need for separate pk certificates. Instead, the DO's sk is derived directly from their identity information, simplifying key management and certificate handling.
- *CLC-based method*: Al-Riyami [54] first proposed the idea of the CLC technique. In this approach, the DO's sk is split into two parts. The key generation center (KGC) generates a partial sk based on the DO's identity and securely sends it to the

DO. The DO then combines it with their own secret information to form a complete sk.

- *ABC-based method*: To enable fine-grained access control, the ABC method has been introduced by Maji et al. [55].

In this approach, the DO selects a specific set of attributes (e.g., name, age, etc.) and submits them to a trusted authority (TA) to obtain a sk.

3.2 | Evaluation Criteria

To enable a systematic comparison of existing LBPA schemes, we define evaluation criteria focusing on both security and performance aspects. The security evaluation considers various adversarial models that capture potential threats in real-world cloud environments, including both classical and quantum-resistant scenarios. Therefore, before presenting the comparative analysis, we outline the underlying adversarial models considered in LBPA schemes.

3.2.1 | Adversarial Model

Most existing LBPA schemes consider two basic adversaries—the malicious CSP and the curious TPA. However, real-world cloud environments are prone to more sophisticated and adaptive threats. This subsection extends the discussion to include additional adversarial models such as adaptive adversaries, insider collusion attacks, and quantum side-channel attacks.

- *Malicious CSP*: A malicious CSP may intentionally delete, alter, or manipulate outsourced data while still attempting to generate seemingly valid proofs of data possession to mislead the TPA and conceal data corruption or loss. Such adversaries aim to save storage costs, avoid detection, or compromise the integrity of the DO information. This behavior directly violates the trust and reliability expected in cloud storage services. Therefore, an essential design goal of LBPA schemes is to ensure *public verifiability* and *soundness*, enabling any verifier to detect even minimal data modifications or deletions by a malicious CSP without relying on the DO's secret information.
- *Curious TPA*: A curious TPA follows the auditing protocol correctly but attempts to infer sensitive information about the outsourced data or user identity from the verification process. This semi-honest adversarial behavior highlights the importance of privacy-preserving auditing mechanisms that protect data confidentiality while enabling public verifiability.
- *Adaptive adversaries*: Adaptive adversaries can modify their attack strategy based on previously observed auditing sessions or responses. They analyze prior interactions to adaptively choose which data blocks, keys, or protocol parameters to target in future attacks. Addressing adaptive adversaries requires schemes that maintain security under multiple and dynamic protocol executions.
- *Insider collusion attacks*: In this scenario, an insider entity—such as the CSP, TPA, or even a compromised DO—may

collude to forge auditing results, leak secret information, or manipulate proofs. Collusion-resistant auditing mechanisms ensure that no subset of entities can compromise the correctness or privacy of the verification process.

- *Quantum side-channel attacks*: Beyond algorithmic quantum attacks, quantum side-channel adversaries exploit implementation level leakages such as timing, power consumption, or electromagnetic emissions—from post-quantum cryptographic operations to recover secret keys.

3.2.2 | Security Evaluation Metrics

We compare selected LBPA schemes based on key security features, including public verifiability, soundness, storage correctness, data dynamics, privacy preservation, quantum resistance, the certificate management problem (CMP), and the key escrow problem (KEP).

- *Public verifiability*: Permits any authorized third party to validate the integrity of outsourced data without requiring access to the underlying secret data.
- *Soundness*: Ensures that a malicious CSP cannot forge a valid proof of data integrity. If data has been tampered with, it will be detected during verification.
- *Storage correctness*: Confirms that CSPs are faithfully storing the DO's data, ensuring no unauthorized modification, deletion, or corruption.
- *Data dynamics*: Enables the DO to flexibly perform operations such as insertion, modification, and deletion on outsourced data blocks at any time.
- *Privacy-preserving*: Ensures that the TPA cannot obtain any sensitive information about the DO or the outsourced data during the auditing process.
- *Quantum-resistance*: Ensures that the cryptographic protocol remains secure against adversaries equipped with quantum computational power.
- *CMP*: In PKI-based systems, public keys require certificates issued by a CA, which leads to significant computational overhead due to certificate generation, renewal, and revocation processes.
- *KEP*: In IBC-based systems, the KGC generates secret keys for DOs, which may allow the KGC to misuse or disclose keys to unauthorized parties.
- *Hardness assumption*: LBPA schemes are typically constructed based on lattice hard problems, which ensure security under worst-case assumptions.

3.2.3 | Performance Evaluation

We assess the performance by analyzing computation and storage costs.

1. *Computation cost*: This refers to the computational resources required for the following stages:

- *SignGen*: The computational cost incurred by the DO for generating a signature for every data block of the data file.
- *ProofGen*: The cost involved in generating a proof by the CSP in reply to a challenge raised by the TPA.
- *ProofVerify*: The cost for the TPA to verify the proof of data possession provided by the CSP.

2. *Storage cost*: The storage cost mainly depends on the representation size of the public key, secret key, and signature, each typically expressed in terms of lattice dimensions n , m , and the modulus q . The bit length required to store elements in $\mathbb{Z}_q$ is denoted as $\log q$.
 - *Public key size*: The storage cost required for the DO's public key.
 - *Secret key size*: The storage cost required for the DO's secret key.
 - *Signature size*: The storage cost required for the signature of each data block of the DO's data file.

4 | Detailed Description and Evaluation of LBPA Schemes

4.1 | PKI-Based LBPA Schemes

By using the PKI method, several LBPA schemes [17–24] are designed. Here, we present the architecture, description, and performance of the above schemes.

4.1.1 | Architecture

The PKI-based LBPA architecture consists of three participants, as depicted in Figure 3.

- *DO*: It is an entity that wants to store its massive data in the cloud due to the limited local storage space.
- *CSP*: It is an entity with unlimited computing power, storage capacity, and responsibility for maintaining the DO's outsourced data.

- *TPA*: It is an independent entity that is honest but curious. It performs the data integrity verification at the DO's request on behalf of the DO without getting any content from the data file.

4.1.2 | Description of PKI-Based LBPA Schemes

This section presents a detailed description of existing PKI-based LBPA schemes built upon SIS (ISIS) and R-LWE assumptions.

- *PKI-based LBPA schemes from SIS (ISIS)*: Liu et al. [17] presented the first LBPA scheme for cloud storage, establishing the foundation for post-quantum data integrity verification. Their model enables a TPA to verify the correctness of outsourced data using homomorphic signatures without retrieving the entire file. The construction ensures efficient verification and reduces communication costs through the aggregation of verification tags. However, the design assumes an honest cloud server and lacks a rigorous security model against malicious entities. It also does not provide mechanisms for data dynamics, which restricts its suitability for real-world cloud environments that frequently involve data updates.

To enhance data privacy, Zhao et al. [18] developed a cloud data integrity checking protocol that incorporates a blind checking mechanism to protect the DO's information from exposure to the TPA. By introducing randomness into the verification process, the scheme preserves privacy but remains vulnerable to collusion attacks between the CSP and TPA, leading to possible biased verification results. Additionally, it incurs high computational overhead during the verification phase due to repeated lattice operations.

Later, Zhang et al. [19] examined Liu et al.'s design and identified critical vulnerabilities. They showed that a dishonest CSP could generate valid proofs without storing the actual data, thus compromising the integrity verification process. This revealed the necessity for stronger binding mechanisms between data

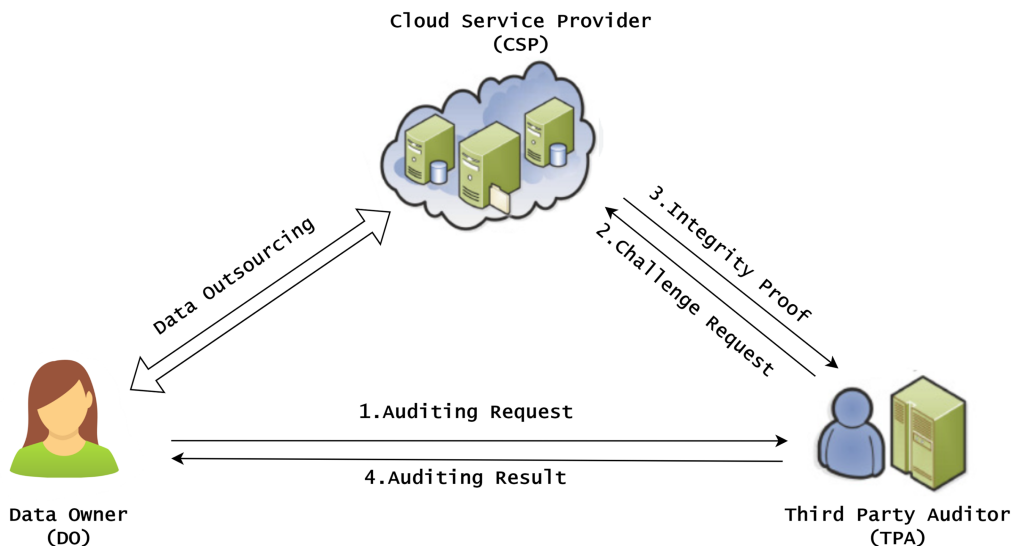


FIGURE 3 | System Model for PKI-based LBPA.

blocks and verification tags, as well as the inclusion of soundness proofs.

To address data update limitations, Yan et al. [20] proposed a dynamic integrity verification scheme that integrates Bloom Filters (BF) for efficient block insertion and membership checking. This approach improves flexibility, but it faces a notable drawback: once a block is added to the BF, it cannot be deleted or modified without affecting the overall accuracy, resulting in potential inconsistencies during long-term auditing.

Further extending this idea, Zhao et al. [21] introduced a dynamic integrity verification framework using a Merkle Hash Tree (MHT) to manage data operations more efficiently. Their scheme adopts a fine-grained signature mechanism to minimize redundant computations and achieve better scalability. However, the linear growth of computational costs with increasing data size continues to affect overall performance, especially in large-scale cloud systems.

Zhou et al. [22] designed a provable data possession (PDP) framework tailored for smart grid data management, emphasizing integrity assurance and verifiability in specialized application scenarios. The scheme enhances security and data traceability but lacks support for batch auditing and multi-user settings, making it less practical for generalized cloud storage models.

To overcome centralization issues, Li et al. [23] proposed a privacy-preserving PDP protocol that employs smart contracts to decentralize the auditing process. This integration of blockchain technology enhances transparency and trust while reducing dependency on a single auditor. Nevertheless, the scheme introduces latency and on-chain storage overhead, which can limit scalability in real-world large-scale applications.

- *PKI-based LBPA scheme from R-LWE*: To reduce computational overhead, Yang et al. [24] designed a secure R-LWE-based scheme that uses vector variables instead of large matrices, thereby improving computational efficiency while maintaining data integrity verification accuracy. This design demonstrates that compact vectorized

representations can significantly enhance performance, making R-LWE-based approaches more practical for large-scale cloud environments.

4.2 | Evaluation of PKI-Based LBPA Schemes

Here, we compare the PKI-based LBPA schemes based on security features and performance criteria.

4.2.1 | Security Analysis

A fine-grained assessment of the PKI-based LBPA schemes in Table 1 demonstrates that, although most works [17–24] support PV, SD, SC, PP, and QR, they operate under different and sometimes incomplete security models. Several schemes evaluate security in simplified settings where adversaries cannot dynamically corrupt DOs or the cloud server, whereas other works consider stronger models with adaptive queries and malicious CSP behavior. However, even these enhanced models often fail to capture practical threats such as rollback attacks, version manipulation, or key exposure over long-term storage. Moreover, only a few schemes provide truly robust support for DD, and their protection against fine-grained update attacks—such as unauthorized modification, block-index manipulation, or tampering with version history—remains limited.

In addition, although all PKI-based LBPA schemes achieve QR through SIS/ISIS assumptions, they generally overlook algebraic structural attacks, hybrid attacks, and parameter-reuse vulnerabilities, which may arise in repeated auditing sessions. The most significant limitation remains the inherent CMP. All PKI-based schemes rely heavily on a trusted CA for key issuance, renewal, revocation, and validation. This dependence introduces high computational and storage overhead and creates a single point of failure: if the CA is compromised, an adversary can forge certificates or impersonate DOs, undermining the entire auditing framework. These limitations highlight the need for more robust trust models, motivating the IBC-based LBPA schemes discussed in the subsequent sections.

TABLE 1 | Security feature analysis of selected PKI-based LBPA schemes.

Scheme	HA	SM	PV	SD	SC	DD	PP	QR	CMP	KEP
Liu et al. [17]	SIS	ROM	✓	✓	✓	✗	✓	✓	✓	—
Zhao et al. [18]	ISIS	ROM	✓	✓	✓	✓	✓	✓	✓	—
Zhang et al. [19]	SIS	ROM	✓	✓	✓	✗	✓	✓	✓	—
Yan et al. [20]	SIS	ROM	✓	✓	✓	✓	✓	✓	✓	—
Zhao et al. [21]	SIS	ROM	✓	✓	✓	✓	✓	✓	✓	—
Zhou et al. [22]	SIS	St.M	✓	✓	✓	✗	✓	✓	✓	—
Li et al. [23]	ISIS	ROM	✓	✓	✓	✗	✓	✓	✓	—
Yang et al. [24]	R-LWE	St.M	✓	✓	✓	✓	✓	✓	✓	—

Abbreviations: ✓, exists; ✗, does not exist; —, not applicable; CMP, certificate management problem; DD, data dynamics; HA, hardness assumption; KEP, key escrow problem; PP, privacy preserving; PV, public verifiability; ROM, random oracle model; QR, quantum resistance; SC, storage correctness; SD, soundness; SM, security model; St.M, standard model.

4.2.2 | Performance Analysis

Here we present the performance analysis with respect to computation and storage costs.

1. *Computation cost*: The computation costs of various PKI-based LBPA schemes are presented in Table 2. From Table 2 we observe that.
 - *SignGen*: Liu et al. [17] and Zhang et al. [19] each utilize n hash operations and l signature operations. Zhao et al. [18] perform one vector multiplication. Yan et al. [20] employ one signature and l bloom filter sampling operations. Zhao et al. [21] perform one signature, h hash operations, and l sampling operations. Zhou et al. [22] use matrix operations with $m^2 + 3mn$ basic computations. Li et al. [23] perform one signature and one sampling operation. Yang et al. [24] conduct $n\lambda^2$ multiplications plus l sampling operations.
 - *ProofGen*: Hongwei et al. [17] and Zhang et al. [19] each perform $(c + n)$ hash operations. Zhao et al. [18] utilizes one vector multiplication. Yan et al. [20] use c bloom filter samplings plus one vector multiplication. Zhao et al. [21] execute c hash operations and one vector multiplication. Zhou et al. [22] use matrix computations totaling $m^2 + 2(n + c)m$. Li et al. [23] conduct c hash operations and one vector multiplication. Yang et al. [24] carry out $n\eta\lambda^2$ multiplications.
 - *ProofVerify*: Liu et al. [17] and Zhang et al. [19] utilize $(c + n + m)$ multiplications. Zhao et al. [18] conduct two vector multiplications. Yan et al. [20] perform one vector multiplication and one bloom filter sampling. Zhao et al. [21] execute one vector multiplication and one signature verification. Zhou et al. [22] use matrix operations totaling $(c + 1)(m + 1)n + nm$. Li et al. [23] perform one vector multiplication and one sampling. Yang et al. [24] execute $n\eta\lambda^2$ multiplications.
2. *Storage cost*: This section presents the storage costs of selected PKI-based schemes in Table 3. From Table 3 we knew that.
 - *Public key size*: Liu et al. [17], Zhang et al. [19], Yan et al. [20], and Zhao et al. [21] all maintain compact public keys

of size $O(n \log q)$. Zhao et al. [18] and Zhou et al. [22] employs a more complex matrix-based design, resulting in a larger public key size of $O(mn \log q)$. The scheme by Li et al. [23] uses of size $O(m \log q)$. In contrast, the R-LWE-based scheme by Yang et al. [24] incorporates the polynomial degree λ , leading to a public key size of $O(n\lambda \log q)$.

- *Secret key size*: Liu et al. [17], Zhang et al. [19], Yan et al. [20], Zhao et al. [21], and Li et al. [23], the secret key size is $O(m \log q)$. Zhao et al. [18] and Zhou et al. [22] use more extensive matrix structures, which increases the secret key size to $O(mn \log q)$. While Yang et al. [24]'s R-LWE-based scheme results in a secret key size of $O(n\lambda \log q)$.
- *Signature size*: Liu et al. [17], Zhang et al. [19], Yan et al. [20], Zhao et al. [21] and Li et al. [23] all produce signatures of size $O(l \log q)$, where l is the number of data blocks. In contrast, Zhao et al. [18] optimizes the signature size down to $O(m \log q)$, despite having large key sizes. Zhou et al. [22] results in significantly larger signatures with size $O(mn \log q)$, matching the sizes of its keys due to the use of large matrices. Finally, Yang et al. [24]'s R-LWE-based approach leads to signature sizes of $O(l\lambda \log q)$.

TABLE 3 | Performance analysis (storage cost) of selected PKI-based LBPA schemes.

Scheme	Public key size	Secret key size	Signature size
Liu et al. [17]	$O(n \log q)$	$O(m \log q)$	$O(l \log q)$
Zhao et al. [18]	$O(mn \log q)$	$O(mn \log q)$	$O(m \log q)$
Zhang et al. [19]	$O(n \log q)$	$O(m \log q)$	$O(l \log q)$
Yan et al. [20]	$O(n \log q)$	$O(m \log q)$	$O(l \log q)$
Zhao et al. [21]	$O(n \log q)$	$O(m \log q)$	$O(l \log q)$
Zhou et al. [22]	$O(mn \log q)$	$O(mn \log q)$	$O(mn \log q)$
Li et al. [23]	$O(m \log q)$	$O(m \log q)$	$O(l \log q)$
Yang et al. [24]	$O(n\lambda \log q)$	$O(n\lambda \log q)$	$O(l\lambda \log q)$

TABLE 2 | Performance analysis (computation cost) of selected PKI-based LBPA schemes.

Scheme	HAS	SignGen	ProofGen	ProofVerify
Liu et al. [17]	SIS	$nT_h + lT_{\text{sig}}$	$(c + n)T_h$	$(c + n + m)T_{\text{mul}}$
Zhao et al. [18]	ISIS	T_{spr}	T_{mv}	$2T_{\text{mv}}$
Zhang et al. [19]	SIS	$nT_h + lT_{\text{sig}}$	$(c + n)T_h$	$(c + n + m)T_{\text{mul}}$
Yan et al. [20]	SIS	$T_{\text{sig}} + lT_{\text{bf}}$	$cT_{\text{bf}} + T_{\text{mv}}$	$T_{\text{mv}} + T_{\text{bf}}$
Zhao et al. [21]	SIS	$T_{\text{sig}} + hT_h + lT_{\text{spr}}$	$cT_h + T_{\text{mv}}$	$T_{\text{mv}} + T_{\text{sig}}$
Zhou et al. [22]	SIS	$m^2 + 3mn$	$m^2 + 2(n + c)m$	$(c + 1)(m + 1)n + nm$
Li et al. [23]	ISIS	$T_{\text{sig}} + T_{\text{sc}}$	$cT_h + T_{\text{mv}}$	$T_{\text{mv}} + T_{\text{sc}}$
Yang et al. [24]	R-LWE	$n\lambda^2 + lT_{\text{spr}}$	$n\eta\lambda^2$	$n\eta\lambda^2$

Note: The performance metrics reported in this table (e.g., signing time, proof generation, and verification costs) are taken from the respective source papers. Empirical simulations will be explored in future work to validate practical feasibility.

Abbreviations: λ , security parameter; c , number of challenged data blocks; HAS, hardness assumption for security; l , number of data blocks; n, m, h , matrix/vector dimensions; T_{sc} , smart contract execution time; T_{bf} , bloom filter operation time; T_h , hash operation time; T_{mv} , matrix-vector multiplication; T_{sig} , signature generation time; T_{spr} , time cost for SamplePre.

4.3 | IBC-Based LBPA Schemes

The LBPA schemes [25–37] designed using the IBC-based method. Here, we provide architecture, and descriptions of those schemes.

4.3.1 | Architecture

The IBC-based LBPA schemes consist of four entities, as represented in Figure 4; in addition to the entities specified in the PKI-based LBPA scheme architecture (DO, CSP, and TPA), there is one more entity, such as KGC.

- **KGC:** It is an entity responsible for generating the public-secret key pair for DO with the DO's identity [16].

4.3.2 | Description of IBC-Based LBPA Schemes

Analyses of existing IBC-based LBPA schemes based on SIS (ISIS) and R-LWE assumptions are presented in this section.

- **IBC-based LBPA schemes from SIS (ISIS):** Zhang et al. [25] pioneered an efficient identity-based PA protocol that employed homomorphic linear authenticators with random masking to preserve data privacy against the TPA and integrity against the CSP. However, it lacked dynamic data support and did not prevent proxy misuse.

Building on this, Zhang et al. [26] enhanced the framework by incorporating a preimage sampleable function (PSF) to generate secure signatures. While this advancement improved resistance to forgery, the design assumed honest behavior from all participants and lacked mechanisms for proxy revocation or misbehavior handling. Liu et al. [27] presented one of the earliest RDIC schemes constructed on lattice primitives. The scheme employed a basis delegation algorithm to generate DO sk and support PV by a TPA. However, this model lacked forward security, making it vulnerable to key exposure attacks. Furthermore, it did not consider dynamic data operations or resistance against

malicious auditors, which restricts its adaptability to evolving cloud environments.

To improve audit flexibility and reduce DO computation, Zhang et al. [28] proposed a delegated auditing model (DOPIV) where a proxy, authorized via a signed warrant, assists in performing data verification. This delegation mechanism significantly reduces the DO's workload but introduces dependency on proxy integrity. If the proxy is compromised, the entire verification process becomes unreliable.

To mitigate key leakage issues, Zhang et al. [29] later introduced a forward-secure variant that updates sk across discrete time intervals without increasing key size. Although this reduces exposure risks, it introduces synchronization overhead and management complexity in multi-user cloud settings.

To enhance security guarantees, Li et al. [30] developed a privacy-preserving and forward-secure PA framework that introduced encapsulation, random masking, and key update mechanisms. These features ensured that previously generated keys remained secure even if the current key was compromised. Despite these improvements, the scheme assumes a fully trusted TPA and does not account for potential collusion between the TPA and CSP, which could undermine auditing reliability.

For application-specific integrity verification, Zhang et al. [31] designed a lattice-based designated verifier auditing system (LDVAS) for electronic medical records in WBANs. This model improves privacy and restricts verification rights to authorized medical personnel. However, it requires the secure transmission of sk, which may raise usability and key management concerns in distributed healthcare infrastructures.

Similarly, Wang et al. [32] proposed a proxy-enabled auditing scheme that provides privacy protection and forward security while offloading computational costs from the DO. Although this design effectively balances performance and privacy, it still depends on proxy trustworthiness and lacks formal mechanisms to prevent proxy misbehavior or collusion with the CSP.

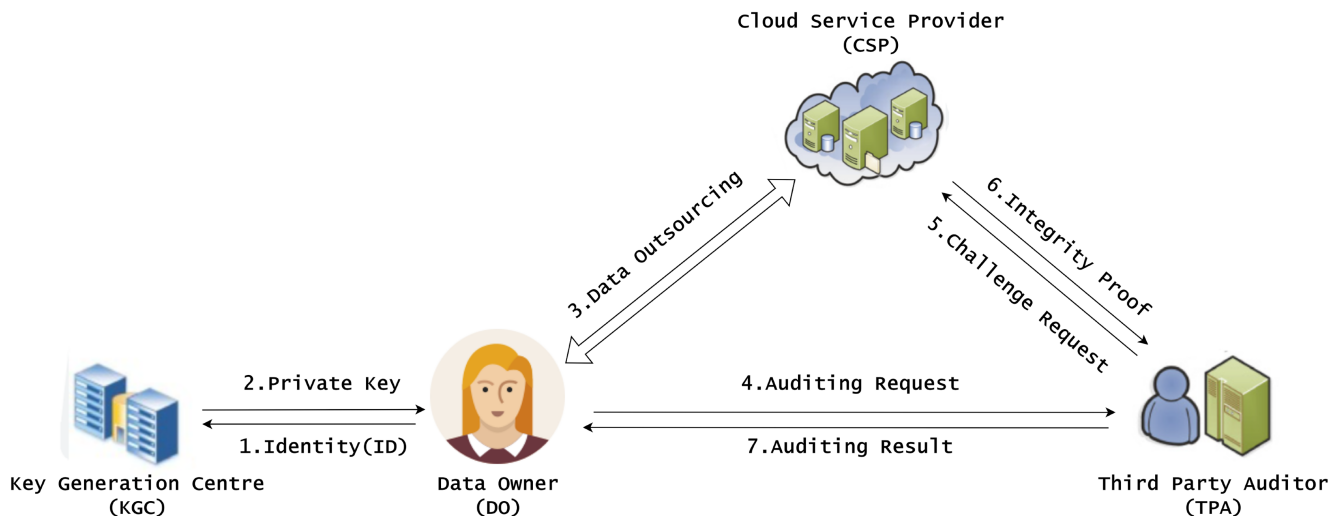


FIGURE 4 | System model for IBC-based LBPA.

In the context of healthcare data, Liu et al. [33] proposed a proxy-oriented auditing mechanism for electronic health records (EHRs) in cloud-assisted wireless body area networks (WBANs). The scheme reduces the computational burden on the DO by allowing a proxy to handle verification tasks and integrates blockchain-based accountability mechanisms. Although this design supports efficient delegation, the additional blockchain layer significantly increases communication overhead and system complexity, making it less practical for lightweight or real-time healthcare systems.

Recognizing vulnerabilities in the DOPIV scheme, Wang et al. [34] analyzed its weaknesses and proposed an enhanced construction resistant to insider threats and proof forgery. Their design strengthens verification correctness but still depends on secure initial key distribution and trusted communication channels, leaving potential exposure to data leakage via compromised proxies.

To address flaws in prior identity-based designs that used encryption instead of signing for tag generation, Yang et al. [35] proposed an improved auditing model for secure cloud storage. This scheme enhances resistance to forgery attacks and strengthens proof correctness. However, it continues to rely on a centralized TPA, which may lead to biased verification results in case of auditor compromise.

To minimize such bias, Gautam et al. [36] introduced a blockchain-assisted scheme designed for multimedia cloud storage. Their model enhances transparency and traceability of verification results using distributed ledger technology. While this architecture eliminates single points of failure, it introduces latency and high computational costs associated with on-chain operations, which limits scalability in high-frequency auditing scenarios.

- *IBC-based LBPA schemes from R-LWE*: Yang et al. [37] proposed three R-LWE-based identity-based data integrity auditing (ID-DIA) protocols that improved efficiency and semantic security compared to earlier lattice-based designs. The schemes utilized ring-based trapdoor generation and efficient sampling algorithms to reduce computation and communication overhead. These designs demonstrate that R-LWE-based constructs can achieve stronger performance while maintaining post-quantum security guarantees. However, they do not support data dynamics or batch auditing, and key generation remains computationally expensive for large-scale datasets, limiting real-world deployment efficiency.

4.4 | Evaluation of IBC-Based LBPA Schemes

This section presents an evaluation of IBC-based LBPA schemes, focusing on their security properties and performance characteristics.

4.4.1 | Security Analysis

A detailed examination of the IBC-based LBPA schemes in Table 4 shows that the schemes [25–37] achieve several core

auditing properties such as PV, SD, SC, PP, and QR. However, these schemes are built upon different levels of adversarial strength and security assumptions, leading to variations in how well they handle real-world threats. Most existing works analyze security under restricted models where the cloud server is only partially malicious or where DO corruption is not fully adaptive. Only [26] provides support for dynamic DD, yet even this scheme does not comprehensively address risks such as rollback manipulation, unauthorized block-index changes, or tampering with version histories. This indicates that fine-grained dynamic threat protection in IBC-based auditing remains limited.

Despite eliminating CMP and providing QR, all IBC-based LBPA schemes inherently suffer from the KEP. Since the KGC is responsible for generating or deriving the DO's private key, it possesses full knowledge of the DO's signing key. This creates a powerful insider threat: a malicious or compromised KGC can impersonate DOs, forge signatures, manipulate stored data, or collude with the CSP without detection. Moreover, none of the current schemes provide mechanisms to mitigate KGC misuse, model malicious KGC behavior, or defend against adaptive identity-based attacks. These limitations highlight the structural weakness of IBC-based auditing and motivate the transition toward CLC-based LBPA schemes, which aim to reduce or eliminate trust in the key-generation authority.

4.4.2 | Performance Analysis

This section highlights the performance with respect to computation and storage costs.

1. *Computation cost*: The computation cost of numerous IBC-based LBPA schemes is structured in Table 5. From this table we observe that,
 - *SignGen*: Zhang et al. [25] use one hash and nm multiplications. Zhang et al. [26] involve $(n + l)$ hashes, $(nml + n^2l)$ multiplications, one trapdoor sampling, and one signature operation. Liu et al. [27] employ n hash operations, n multiplications, and l trapdoor sampling operations. Zhang et al. [28] use 2 hashes, $(n + m)$ multiplications, and one trapdoor sampling. The scheme by Zhang et al. [29] requires 3 hashes, $(n + ml)$ multiplications, and one trapdoor sampling. Li et al. [30] use l hashes, $(2nml + n^2l)$ multiplications, one trapdoor sampling, and one signature operation. Zhang et al. [31] perform 2 hashes and mn multiplications. Wang et al. [32] execute $2n$ hashes and $(n + 2nm)$ multiplications. Liu et al. [33] perform one trapdoor sampling only. Wang et al. [34] use 3 hashes and nm multiplications. Yang et al. [35] utilize n hashes, n multiplications, and l trapdoor samplings. Gautam et al. [36] conduct $3n^2$ multiplications. Yang et al. [37] carry out $(n\lambda^2 + T_{spr})l$ operations, where λ is the ring dimension parameter.
 - *ProofGen*: Zhang et al. [25] perform $(2cm + 1)$ hashes. Zhang et al. [26] execute one hash, m multiplications, and one trapdoor sampling. Liu et al. [27] perform one trapdoor sampling and one hash. Zhang et al. [28] and Zhang et al. [29] perform $(2cm + 1)$ and $(2cm + 2)$ hash, respectively. Li et al. [30] conduct $(nm + 2n + mc)$ multiplications and one trapdoor sampling. Zhang et al. [31]

TABLE 4 | Security evaluation of IBC-based LBPA schemes.

Scheme	HA	SM	PV	SD	SC	DD	PP	QR	CMP	KEP
Zhang et al. [25]	ISIS	ROM	✓	✓	✓	✗	✓	✓	✗	✓
Zhang et al. [26]	ISIS	ROM	✓	✓	✓	✓	✓	✓	✗	✓
Liu et al. [27]	SIS	ROM	✓	✓	✓	✗	✓	✓	✗	✓
Zhang et al. [28]	ISIS	ROM	✓	✓	✓	✗	✓	✓	✗	✓
Zhang et al. [29]	ISIS	ROM	✓	✓	✓	✗	✓	✓	✗	✓
Li et al. [30]	SIS	ROM	✓	✓	✓	✗	✓	✓	✗	✓
Zhang et al. [31]	ISIS	ROM	✓	✓	✓	✗	✓	✓	✗	✓
Wang et al. [32]	ISIS	ROM	✓	✓	✓	✗	✓	✓	✗	✓
Wang et al. [34]	ISIS	ROM	✓	✓	✓	✗	✓	✓	✗	✓
Liu et al. [33]	SIS	ROM	✓	✓	✓	✗	✓	✓	✗	✓
Yang et al. [35]	SIS	ROM	✓	✓	✓	✗	✓	✓	✗	✓
Gautam et al. [36]	SIS	ROM	✓	✓	✓	✗	✓	✓	✗	✓
Yang et al. [37]	R-LWE	St.M	✓	✓	✓	✗	✓	✓	✗	✓

Abbreviations: ✓, exists; ✗, does not exist; –, not applicable; CMP, certificate management problem; DD, data dynamics; HA, hardness assumption; KEP, key escrow problem; PP, privacy preserving; PV, public verifiability; ROM, random oracle model; QR, quantum resistance; SC, storage correctness; SD, soundness; SM, security model; St.M, standard model.

TABLE 5 | Performance analysis (computation cost) of selected IBC-based LBPA schemes.

Scheme	HAS	SignGen	ProofGen	ProofVerify
Zhang et al. [25]	ISIS	$T_h + nmT_{mul}$	$(2cm + 1)T_h$	$(n + c + 1)T_h + (n^2 + 2nm + n)T_{mul}$
Zhang et al. [26]	ISIS	$(n + l)T_h + (nml + n^2l)T_{mul} + T_{spr} + T_{sig}$	$T_h + mT_{mul} + T_{spr}$	$(n + c + 1)T_h + (cn + 2mn + n^2 + 2n)T_{mul}$
Liu et al. [27]	SIS	$nT_h + nT_{mul} + lT_{spr}$	$T_{spr} + T_h$	$(n + c)T_h + 3T_{mul}$
Zhang et al. [28]	ISIS	$2T_h + (n + m)T_{mul} + T_{spr}$	$(2cm + 1)T_h$	$(n + c + 2)T_h + (n^2 + 2nm + n)T_{mul}$
Zhang et al. [29]	ISIS	$3T_h + (n + ml)T_{mul} + T_{spr}$	$(2cm + 2)T_h$	$(n + c + 2)T_h + (2nm + cn + n^2)T_{mul}$
Li et al. [30]	SIS	$lT_h + (2nml + n^2l)T_{mul} + T_{spr} + T_{sig}$	$(nm + 2n + mc)T_{mul} + T_{spr}$	$cT_h + (2nm + nc)T_{mul}$
Zhang et al. [31]	ISIS	$2T_h + mnT_{mul}$	$(2cm + m + mn)T_{mul} + 2T_h$	$(n + c + 1)2T_h + (c + cn + 2n + mn)T_{mul}$
Wang et al. [32]	ISIS	$2nT_h + (n + 2nm)T_{mul}$	$(2cm + 1)T_h$	$(n + c + 1)T_h + (n^2 + 2nm + n)T_{mul}$
Liu et al. [33]	SIS	T_{spr}	$cT_h + (m + n)cT_{mul}$	$cT_h + (2mn + nc)T_{mul}$
Wang et al. [34]	ISIS	$3T_h + nmT_{mul}$	$(3cm + 1)T_h$	$(n + c + 2)T_h + (n^2 + 3nm + n)T_{mul}$
Yang et al. [35]	SIS	$nT_h + nT_{mul} + lT_{spr}$	$T_{spr} + cT_h$	$(n + c + 1)T_h + 3T_{mul}$
Gautam et al. [36]	SIS	$3n^2$	$2cmn$	mn^2
Yang et al. [37]	R-LWE	$(n\lambda^2 + T_{spr})l$	$\eta n\lambda^2 + T_{spr}$	$\eta n\lambda^2$
• (2nd protocol)	R-LWE	$(n\lambda^2 + T_{spr})l$	$\eta n\lambda^2$	$\eta n\lambda^2$
• (3rd protocol)	R-LWE	$(n\lambda^2 + T_{spr})l$	$\eta n\lambda^2$	$\eta n\lambda^2$

Note: The performance metrics reported in this table (e.g., signing time, proof generation, and verification costs) are taken from the respective source papers. Empirical simulations will be explored in future work to validate practical feasibility.

Abbreviations: λ , security parameter; c , number of challenged data blocks; HAS, hardness assumption for security; l , number of data blocks; n, m, h , matrix/vector dimensions; T_{sc} , smart contract execution time; T_{bf} , bloom filter operation time; T_h , hash operation time; T_{mv} , matrix–vector multiplication; T_{sig} , signature generation time; T_{spr} , time cost for SamplePre.

perform 2 hashes and $(2cm + m + mn)$ multiplications. Wang et al. [32] use $(2cm + 1)$ hashes. Liu et al. [33] use c hashes and $(m + n)c$ multiplications. Wang et al. [34] conduct $(3cm + 1)$ hashes. Yang et al. [35] use one trapdoor sampling and c hashes. Gautam et al. [36] execute $2cmn$ multiplications. Yang et al. [37] conduct $\eta n \lambda^2 + T_{spr}$ operations.

- **ProofVerify:** Zhang et al. [25] use $(n + c + 1)$ hashes and $(n^2 + 2nm + n)$ multiplications. Zhang et al. [26] perform $(n + c + 1)$ hashes and $(cn + 2mn + n^2 + 2n)$ multiplications. Liu et al. [27] perform $(n + c)$ hashes and 3 multiplications. Zhang et al. [28] and Zhang et al. [29] use $(n + c + 2)$ hashes with $(n^2 + 2nm + n)$ and $(2nm + cn + n^2)$ multiplications, respectively. Li et al. [30] conduct c hashes and $(2nm + nc)$ multiplications. Zhang et al. [31] use $2(n + c + 1)$ hashes and $(c + cn + 2n + mn)$ multiplications. Wang et al. [32] conduct $(n + c + 1)$ hashes and $(n^2 + 2nm + n)$ multiplications. Liu et al. [33] use c hashes and $(2mn + nc)$ multiplications. Wang et al. [34] perform $(n + c + 2)$ hashes and $(n^2 + 3nm + n)$ multiplications. Yang et al. [35] perform $(n + c + 1)$ hashes and 3 multiplications. Gautam et al. [36] perform mn^2 multiplications. Yang et al. [37] perform $\eta n \lambda^2$ multiplications in all three protocol variants.

2. **Storage cost:** In this section, we examine the storage requirements of IBC-based LBPA schemes, as outlined in Table 6.

- **Public key size:** Liu et al. [27], Yang et al. [35] and Wang et al. [32] utilize efficient constructions with public keys of size $O(n \log q)$. In contrast, the schemes by Li et al. [30] and Zhang et al. [26, 28, 29, 34] require significantly larger public key sizes on the order of $O(nml \log q)$. The scheme by Gautam et al. [36] incorporates quadratic terms in n , leading to increased public key sizes due to its matrix-based and blockchain-integrated construction. R-LWE-based schemes such as Yang et al. [37] define

public key sizes parameterized by the polynomial degree λ , resulting in $O(n \lambda \log q)$ complexity.

- **Secret key size:** Liu et al. [27], Yang et al. [35], Li et al. [30], Liu et al. [33], Zhang et al. [25], Zhang et al. [26], Zhang et al. [28], Zhang et al. [29], Wang et al. [34] and Zhang et al. [31] maintain secret keys of size $O(m \log q)$, keeping them lightweight and efficient. On the other hand, Gautam et al. [36] and Wang et al. [32] may involve secret key sizes as large as $O(nml \log q)$, particularly due to additional delegation or revocation functionalities. R-LWE-based constructions like Yang et al. [37] scale secret key size as $O(n \lambda \log q)$, in line with their ring-based designs.
- **Signature size:** Liu et al. [27], Yang et al. [35], Li et al. [30], Liu et al. [33], Zhang et al. [26], Zhang et al. [28], Zhang et al. [29], Wang et al. [34] and Wang et al. [32] generate signatures of size $O(l \log q)$. Conversely, schemes such as Gautam et al. [36] and Zhang et al. [25] reach sizes of $O(mn \log q)$ and $O((n^2 + nm) \log q)$, driven by more extensive tag and metadata inclusion. Zhang et al. [31] consumed of size $O((c + cn + mn) \log q)$. R-LWE-based solutions like that of Yang et al. [37] produce signatures sized $O(l \lambda \log q)$.

4.5 | CLC-Based LBPA Schemes

This section outlines the architecture and descriptions of selected CLC-based LBPA schemes [38–43].

4.5.1 | Architecture

CLC-based LBPA schemes involve four entities, similar to IBC-based LBPA schemes: the DO, CSP, TPA, and KGC. However, the roles of the DO and KGC in CLC-based LBPA schemes differ from those in IBC-based LBPA schemes. Figure 5 illustrates the architecture of a typical CLC-based LBPA schemes.

4.5.2 | Description of CLC-Based LBPA Schemes

This section explores the existing CLC-based LBPA schemes that are based on SIS (ISIS) and R-LWE assumptions.

- **CLC-based LBPA schemes from SIS (ISIS):** Basha et al. [38] proposed one of the earliest certificateless LBPA schemes, eliminating the dependence on traditional certificates and mitigating key escrow issues inherent in identity-based constructions. Their protocol employs one-way trapdoor functions and lattice basis delegation to generate verification tags for individual data blocks. However, the design lacks resistance against malicious auditors and semi-honest CSPs, allowing possible proof forgery under adaptive auditing scenarios.

To enhance auditing efficiency, Sasikala et al. [39] developed a batch verification framework enabling TPAs to verify multiple DOS' files simultaneously in multi-cloud environments. The scheme leverages Fibonacci-based encoding to detect altered or missing data blocks efficiently, but its static key structure and

TABLE 6 | Performance analysis (storage cost) of selected IBC-based LBPA schemes.

Scheme	Public key size	Secret key size	Signature size
Zhang et al. [25]	$O(nm \log q)$	$O(m \log q)$	$O((n^2 + nm) \log q)$
Zhang et al. [26]	$O(nml \log q)$	$O(m \log q)$	$O(l \log q)$
Liu et al. [27]	$O(n \log q)$	$O(m \log q)$	$O(l \log q)$
Zhang et al. [28]	$O(nml \log q)$	$O(m \log q)$	$O(l \log q)$
Zhang et al. [29]	$O(nml \log q)$	$O(m \log q)$	$O(l \log q)$
Li et al. [30]	$O(nml \log q)$	$O(m \log q)$	$O(l \log q)$
Zhang et al. [31]	$O(mn \log q)$	$O(m \log q)$	$O((c + cn + mn) \log q)$
Wang et al. [32]	$O(n \log q)$	$O(mn \log q)$	$O(l \log q)$
Liu et al. [33]	$O(m \log q)$	$O(m \log q)$	$O(l \log q)$
Wang et al. [34]	$O(nml \log q)$	$O(m \log q)$	$O(l \log q)$
Yang et al. [35]	$O(n \log q)$	$O(m \log q)$	$O(l \log q)$
Gautam et al. [36]	$O(n^2 \log q)$	$O(mn \log q)$	$O(mn \log q)$
Yang et al. [37] (all protocols)	$O(n \lambda \log q)$	$O(n \lambda \log q)$	$O(l \lambda \log q)$

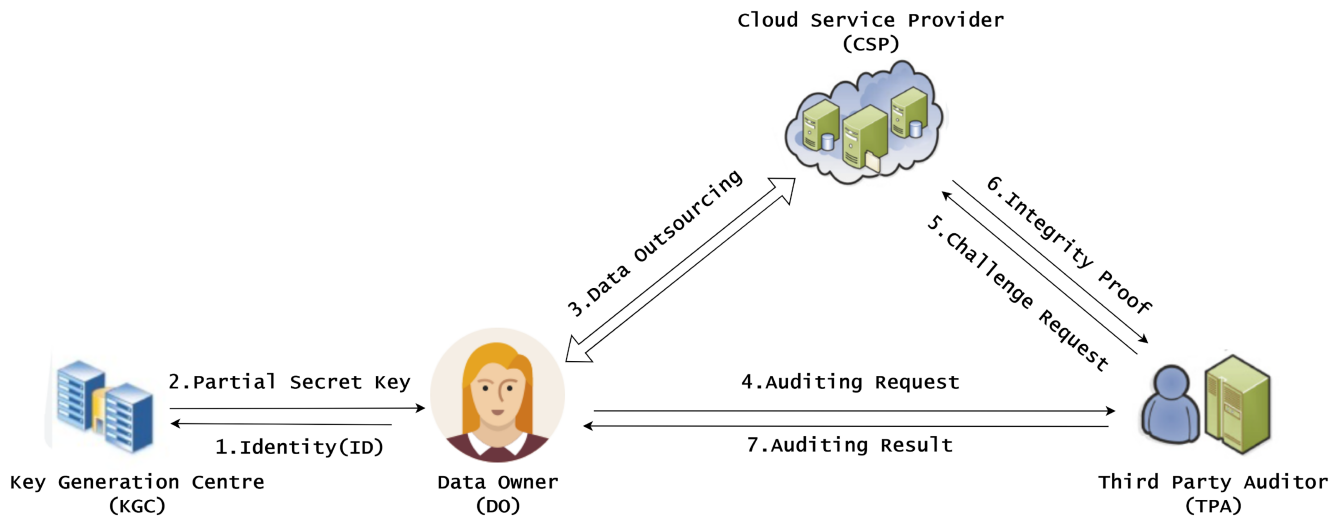


FIGURE 5 | System model for CLC-based LBPA.

TABLE 7 | Security feature analysis of selected CLC-based LBPA schemes.

Scheme	HA	SM	PV	SD	SC	DD	PP	QR	CMP	KEP
Basha et al. [38]	SIS	ROM	✓	✓	✓	✗	✗	✓	✗	✗
Sasikala et al. [39]	SIS	ROM	✓	✓	✓	✗	✓	✓	✗	✗
Sasikala et al. [40]	SIS	ROM	✓	✓	✓	✗	✓	✓	✗	✗
Zhang et al. [41]	SIS	ROM	✓	✓	✓	✗	✓	✓	✗	✗
Li et al. [42]	ISIS	ROM	✓	✓	✓	✗	✓	✓	✗	✗
Renuka et al. [43]	R-LWE	ROM	✓	✓	✓	✗	✓	✓	✗	✗

Abbreviations: ✓, exists; ✗, does not exist; –, not applicable; CMP, certificate management problem; DD, data dynamics; HA, hardness assumption; KEP, key escrow problem; PP, privacy preserving; PV, public verifiability; ROM, random oracle model; QR, quantum resistance; SC, storage correctness; SD, soundness; SM, security model; St.M, standard model.

centralized TPA model limit scalability and resilience to insider threats.

Likewise, Sasikala et al. [40] introduced a lattice-based RDIC mechanism that supports public auditability without requiring a trusted certificate authority. It does not fully protect against replay attacks or data modification by the malicious CSP.

Further, to enhance data redundancy and fault tolerance, Zhang et al. [41] proposed a multi-replica integrity verification protocol that allows a DO to store multiple copies of data across distributed servers within a single cloud while maintaining verifiable consistency. While the scheme strengthens availability and reliability, it introduces significant storage overhead and remains vulnerable to collusion between the CSP and verifier.

Li et al. [42] designed a post-quantum secure certificateless auditing protocol (PSCPAC), integrating multiple optimization techniques—such as lightweight key generation, compressed tag formation, and efficient challenge–response computation—to improve scalability and verification performance. Despite these improvements, the protocol assumes non-colluding TPAs and lacks formal mechanisms to revoke compromised auditing keys, leaving it susceptible to insider forgery attacks.

- *CLC-based LBPA scheme from R-LWE*: To minimize computational overhead, Cheeturi et al. [43] proposed an

R-LWE-based LBPA scheme that employs vector-based variables instead of large integer matrices. Although this approach enhances efficiency, it lacks a mechanism for distributed or zero-knowledge-based verification to ensure complete privacy preservation against curious TPAs in adversarial cloud environments.

4.6 | Evaluation of CLC-Based LBPA Schemes

This section explains the feature and performance analysis of CLC-based LBPA schemes.

4.6.1 | Security Analysis

The assessment of CLC-based LBPA schemes presented in Table 7 shows that all frameworks [38–43] support the essential properties PV, SD, SC, and QR, while PP is addressed only by a subset of schemes [39–42]. These schemes successfully eliminate the CMP found in PKI-based approaches and avoid the KEP inherent in IBC-based systems by dividing the private key into DO-generated and KGC-generated keys. However, the adopted threat models vary significantly. In most works, the adversary is limited to basic data manipulation behaviors, and scenarios involving adaptive DO corruption, CSP–KGC collusion, or adversaries capable of influencing update operations are not fully explored. As a result,

the formal security assurances offered by existing CLC-based schemes differ in strength and completeness.

Furthermore, none of the current CLC-based LBPA schemes demonstrate strong protection against forgery attempts initiated by a malicious CSP. In R-LWE-based schemes, the algebraic structure of authentication tags or the interaction between partial keys may allow a CSP to fabricate valid proofs without holding the correct data blocks. Risks such as manipulated update responses, unauthorized re-indexing of blocks, or exploiting inconsistencies between DO-side and KGC-derived key components remain insufficiently addressed. These limitations reveal that, despite removing CMP and KEP, CLC-based schemes still lack comprehensive defense against malicious CSP adversaries. Strengthening unforgeability guarantees and modeling more realistic CSP behaviors remains an important direction for future improvements.

4.6.2 | Performance Analysis

We evaluate the performance based on computation and storage overhead.

1. *Computation cost*: The computational costs of several CLC-based LBPA schemes are outlined in Table 8. It can be seen that
 - *SignGen*: The scheme by Basha et al. [38] requires n hash operations, l vector multiplications, and l trapdoor sampling operations. The scheme by Sasikala et al. [39] involves l hash computations, l matrix–vector multiplications, and l trapdoor sampling operations. The scheme by Sasikala et al. [40] performs $(n + l)$ hash operations, l multiplications, and l trapdoor samplings. Zhang et al. [41] requires $(n + l)c$ hash computations, lc multiplications, and lc trapdoor samplings. Li et al. [42] executes $(n + l)$ hash operations, $(nml + n^2l)$ multiplications, one trapdoor sampling, and one signature generation. Renuka et al. [43] perform l polynomial multiplications, l hash computations, and l discrete Gaussian samplings.
 - *ProofGen*: The scheme by Basha et al. [38] uses one hash operation and one vector multiplication. Sasikala et al. [39] perform $(c + 1)$ matrix–vector multiplications, c matrix–matrix multiplications, and $(c + 1)$ auxiliary transformations. Sasikala et al. [40] includes up to l multiplications, approximately $l(c + 1)$ trapdoor-related operations, and one hash function. Zhang et al. [41] executes approximately $2lc$ multiplications, $2lc$ trapdoor operations, and c hash computations. Li et al. [42] performs $4m + n$ multiplications. Renuka et al. [43] conduct $2c$ polynomial multiplications.
 - *ProofVerify*: The scheme by Basha et al. [38] performs one hash operation and two vector multiplications. Sasikala et al. [39] includes $c(c + 1)$ matrix–matrix multiplications and $2c$ augmented matrix transformations. The scheme by Sasikala et al. [40] carries out approximately 3 standard multiplications, up to l more multiplications, and nearly lc trapdoor-based computations. Zhang et al. [41] performs $3c$ multiplications and c trapdoor operations. Li et al. [42] executes $(n + c + l)$ hash computations and a total of $(4mn + n^2 + nc + n + 2m)$ multiplications.

TABLE 8 | Performance analysis (computation cost) of selected CLC-based LBPA schemes.

Scheme	HAS	SignGen	ProofGen	ProofVerify
Basha et al. [38]	SIS	$n(h) + ln(mv) + lT_{spr}$	$T_h + T_{mv}$	$T_h + 2T_{mv}$
Sasikala et al. [39]	SIS	$1lh + lT_{mm} + lspr$	$(c + 1)(A + T_{mm}) + cT_{mmm} + A$	$c(c + 1)T_{mmm} + 2cT_{m_{aug}}$
Sasikala et al. [40]	SIS	$(n + l)T_h + lT_{mul} + lT_{spr}$	$(2 \sim l + 1)T_{mul} + (\sim l(c + 1) + 1)T_A + T_{mul} + h_h$	$3T_{mul} + (\sim l + 2)T_{mul} + (\sim lc + 2)T_A$
Zhang et al. [41]	SIS	$(n + l)cT_h + lcT_{mul} + lcT_{spr}$	$(2 \sim l + 1)cT_{mul} + (2 \sim l + 1)cT_A + cT_{mul} + cT_{h_2}$	$3cT_{mul} + cT_A$
Li et al. [42]	ISIS	$(n + l)T_h + (nml + n^2l)T_{mul} + T_{spr} + T_{sig}$	$4m + n$	$(n + c + l)T_h + (4mn + n^2 + nc + n + 2m)T_{mul}$
Renuka et al. [43]	R-LWE	$l(T_{mul_{pol}} + T_h + T_{rspr})$	$2cT_{mul_{pol}}$	$c(T_{mul_{pol}} + T_h)$

Note: rspr — R-LWE SamplePre algorithm. The performance metrics reported in this table (e.g., signing time, proof generation, and verification costs) are taken from the respective source papers. Empirical simulations will be explored in future work to validate practical feasibility.

Abbreviations: λ , security parameter; c , number of challenged data blocks; HAS , hardness assumption for security; l , number of data blocks; n, m, h , matrix/vector dimensions; T_{sc} , smart contract execution time; T_{h} , hash operation time; T_{mv} , matrix–vector multiplication; T_{sig} , signature generation time; T_{spr} , time cost for SamplePre.

Renuka et al. [43] uses c polynomial multiplications and c hash computations.

2. *Storage cost*: Table 9 illustrates the storage costs associated with various CLC-based LBPA schemes. From this data, we analyze that...

- *Public key size*: Basha et al. [38], Sasikala et al. [39, 40], Zhang et al. [41] and Li et al. [42], use matrix-based public keys of dimension $n \times m$, leading to storage costs of $O(nm \log q)$. In contrast, Renuka et al. [43] achieves a more compact representation by encoding public keys as polynomials of degree n , resulting in a storage cost of $O(n \log q)$.
- *Secret key size*: Basha et al. [38], Sasikala et al. [39, 40], and Zhang et al. [41] employ m -dimensional vector keys, yielding storage sizes of $O(m \log q)$. Li et al. [42] uses a larger secret key of dimension $n \times n$ and corresponding storage of $O(n^2 \log q)$. Renuka et al. [43] maintains secret keys as polynomials of degree n , with a compact size of $O(n \log q)$.

- *Signature size*: Basha et al. [38], Sasikala et al. [39, 40], Zhang et al. [41] and Li et al. [42], the signature is a vector of dimension m , leading to a storage cost of $O(m \log q)$. Renuka et al. [43], signatures are polynomial-based and share the compact storage requirement of $O(n \log q)$.

4.7 | ABC-Based LBPA Scheme

This section introduces the architecture and description of the ABC-based LBPA scheme.

4.7.1 | Architecture

As depicted in Figure 6. ABC-based LBPA schemes consist of four entities (TA, DO, CSP, and TPA).

TA: It is responsible for initializing the system by generating the global public parameters and master secret key and for authorizing DOs and auditors by issuing attribute-based private keys according to their assigned attributes.

4.7.2 | Description of ABC-Based LBPA Scheme

- *ABC-based LBPA scheme*: Zhang et al. [44] designed a revocable PA scheme using the SIS lattice hard problem. In this method, the TA generates the DO's secret key using the DO's identity and attribute set, and during sign generation, the DO uses the linear secret sharing scheme (LSSS) to secure fine-grained management of his permission. When the DO resigns or the role is changed, the TA revokes its authorization and reauthorizes the DO. However, this scheme is applicable to only private clouds, and if the TA compromises, the entire system will collapse.

TABLE 9 | Performance analysis (storage cost) of selected CLC-based LBPA schemes.

Scheme	Public key size	Secret key size	Signature size
Basha et al. [38]	$O(nm \log q)$	$O(m \log q)$	$O(m \log q)$
Sasikala et al. [39]	$O(nm \log q)$	$O(m \log q)$	$O(m \log q)$
Sasikala et al. [40]	$O(nm \log q)$	$O(m \log q)$	$O(m \log q)$
Zhang et al. [41]	$O(nm \log q)$	$O(m \log q)$	$O(m \log q)$
Li et al. [42]	$O(nm \log q)$	$O(n^2 \log q)$	$O(m \log q)$
Renuka et al. [43]	$O(n \log q)$	$O(n \log q)$	$O(n \log q)$

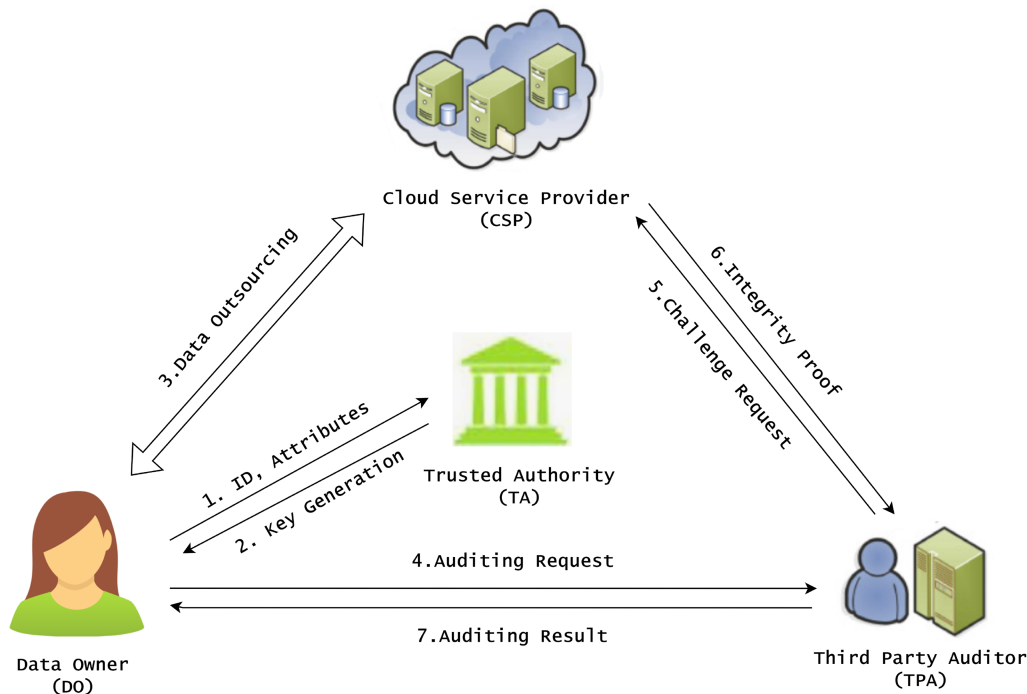


FIGURE 6 | System model for ABC-based LBPA.

TABLE 10 | Security feature analysis of selected ABC-based LBPA schemes.

Scheme	HA	SM	PV	SD	SC	DD	PP	QR	CMP	KEP
Zhang et al. [44]	SIS	ROM	✓	✓	✓	✗	✗	✓	✗	✓

Abbreviations: ✓, exists; ✗, does not exist; –, not applicable; CMP, certificate management problem; DD, data dynamics; HA, hardness assumption; KEP, key escrow problem; PP, privacy preserving; PV, public verifiability; ROM, random oracle model; QR, quantum resistance; SC, storage correctness; SD, soundness; SM, security model; St.M, standard model.

TABLE 11 | Performance analysis (computation cost) of ABC-based LBPA scheme.

Scheme	HAS	SignGen	ProofGen	ProofVerify
Zhang et al. [44]	SIS	$T_h + T_{spr} + (nm_1 + m_1m_2)$	$T_h + (nm_2 + n)$	$T_h + (cn + nm_1 + nm_2 + n)$

Note: All abbreviations are given under Table 2.

TABLE 12 | Performance analysis (storage cost) of ABC-based LBPA scheme.

Scheme	Public key size	Secret key size	Signature size
Zhang et al. [44]	$O(nm_1 \log q + nm_2 \log q)$	$O(m_1^2 \log q)$	$O(m \log q)$

4.8 | Evaluation of ABC-Based LBPA Scheme

This section explains the feature and performance analysis of the ABC-based LBPA scheme.

4.8.1 | Security Analysis

From the security assessment in Table 10, the ABC-based LBPA scheme [44] does not support DD or PP and suffers from KEP, as the central authority generates DO's keys. This allows the TA to potentially reconstruct the DO's key and impersonate the legitimate DO. The scheme also does not model stronger threats such as CSP–TA collusion, attribute manipulation, or attacks arising from outdated attribute keys during revocation. These gaps show that the current ABC-based design offers limited security protection and requires stronger adversarial modeling and improved revocation mechanisms.

4.8.2 | Performance Analysis

This section provides the performance of the ABC-based scheme.

1. **Computation cost:** Table 11 summarizes the computational costs of the ABC-based LBPA scheme. The comparison clearly shows that ...
 - **SignGen:** The signer performs one hash operation (T_h) to compute the identity or attribute hash. A trapdoor sampling operation (T_{spr}) is then used to generate a short vector solution to the lattice equation. The computation includes matrix–vector multiplication with a public matrix $A \in \mathbb{Z}_q^{n \times m_1}$, resulting in a cost of nm_1 , and an additional randomization step with matrix $R \in \mathbb{Z}_q^{m_1 \times m_2}$ contributes a cost of m_1m_2 . Thus, the total cost is $T_h + T_{spr} + nm_1 + m_1m_2$.
 - **ProofGen:** The CSP performs one hash operation (T_h) to generate challenge coefficients. It then multiplies a public matrix $B \in \mathbb{Z}_q^{n \times m_2}$ with a vector, leading to a cost of nm_2 .

A lightweight vector operation across dimension n adds an extra cost of n . Therefore the total computation cost is $T_h + nm_2 + n$.

- **ProofVerify:** The TPA verifies the proof by first performing one hash computation (T_h) to derive the challenge. Then, it carries out several matrix–vector multiplications: (cn) for combining challenge coefficients and response vectors, (nm_1) with matrix A , and (nm_2) with matrix B . A final vector comparison adds a cost of n . Hence, the overall cost becomes $T_h + (cn + nm_1 + nm_2 + n)$.
- 2. **Storage cost:** In this section we display the ABC-based LBPA schemes' storage cost, which is included in Table 12. From Table 12, we analyze that ...
 - **Public key size:** The scheme by Zhang et al. [44] utilizes a public key size of the order $O(nm_1 \log q + nm_2 \log q)$.
 - **Secret key size:** The scheme by Zhang et al. [44] takes the size $O(m_1^2 \log q)$ for the secret key.
 - **Signature size:** The scheme Zhang et al. [44] produces the signature size $O(m \log q)$.

The evolution of LBPA schemes demonstrates a progressive transition from *PKI-based* frameworks toward *identity-based*, *certificateless*, and *attribute-based* constructions, each addressing limitations of earlier models. While recent schemes improve efficiency, privacy, and fine-grained access control, most still depend on centralized auditors and trapdoor mechanisms. Future research should aim to design *decentralized*, *trapdoor-free*, and *quantum-resilient* auditing frameworks that ensure scalable and transparent data integrity verification across multi-cloud environments.

5 | Challenges and Future Directions

From the security features and performance analysis presented in Tables 1–12, several critical issues in existing LBPA schemes have been identified that need to be addressed in future directions.

5.1 | Efficient and Secure LBPA Schemes

Although all LBPA [17–44] schemes guarantee data integrity in the cloud with quantum resistance, they are inefficient and insecure due to the following reasons:

- *Inefficient*: The majority of LBPA schemes [17–23, 25–36, 38–42, 44] are designed based on SIS (ISIS) lattice problems, which are theoretically sound but inefficient in practice due to the use of large, unstructured matrices, as well as matrix–matrix and matrix–vector multiplications. Additionally, all current LBPA schemes adopt the hash-and-sign paradigm, which relies on complex trapdoor generation and preimage sampling algorithms. These create large public/secret key pairs and signature sizes, take a long time to execute, and lead to significant computational and storage overhead [24, 37, 43].
- *Insecure*: To address the limitations of SIS/ISIS-based schemes and enhance efficiency, several researchers have proposed R-LWE-based constructions [24, 37, 43], which rely on ring polynomials. Although these schemes offer performance benefits, they remain susceptible to algebraic attacks [15] that exploit the inherent algebraic structure of the ring. Such attacks can leverage structural weaknesses in the underlying problem and may enable adversaries to recover secret keys or other sensitive information, ultimately undermining the overall security of R-LWE-based LBPA schemes.

5.1.1 | Mitigation Through Module Lattices and Hybrid Assumptions

To mitigate the above vulnerabilities, recent research has focused on extending R-LWE to module-lattice frameworks such as *Module-LWE* and *Module-SIS*, which generalize ring-lattice constructions by embedding the problem in higher-dimensional modules to reduce exploitable algebraic regularities while maintaining computational efficiency.

For example, Mukherjee et al. [56] proposed a modular homomorphic encryption framework (ModHE) based on the M-LWE assumption that provides improved resistance to algebraic and structural attacks. Similarly, Liu et al. [57] enhanced Kyber-like designs through lattice quantization techniques that strengthen the module structure and optimize parameter efficiency. Furthermore, de Boer et al. [58] conducted a comprehensive comparison between R-LWE and M-LWE, concluding that module lattices offer a more favorable trade-off between algebraic soundness and performance. In addition, Sabani et al. [59] explored hybrid hardness assumptions combining M-LWE and SIS, demonstrating improved robustness against cross-structure lattice reductions.

Therefore, adopting module-lattice frameworks and hybrid assumptions in future LBPA schemes can effectively mitigate algebraic vulnerabilities of R-LWE while ensuring stronger post-quantum resilience and operational efficiency.

5.2 | LBPA for Cloud-Assisted IoT

The cloud-assisted Internet of Things (CIoT) is a rapidly evolving technology that utilizes cloud storage to manage the extensive data generated by IoT devices, which typically have limited storage and computing capacity. To ensure the integrity of IoT data in the cloud, it is essential to implement a public auditing protocol. However, existing LBPA schemes [17–44] are not suitable for cloud-assisted IoT devices because they are designed based on lattice-based hash-and-sign signatures that utilize trapdoor sampling algorithms. These algorithms take a long time to execute and increase computational overhead, particularly during the *keygen*, *signgen* and *proofgen* stages.

Therefore, it is necessary to design a lightweight LBPA for cloud-assisted IoT based on Fiat-Shamir lattice-based signatures instead of hash-and-sign signatures. The new lightweight LBPA schemes will enable cloud-assisted IoT devices to handle their data safely and efficiently, overcoming the issues created by existing LBPA schemes.

5.3 | LBPA for Shared Data

In cloud-based shared storage environments, DOs often store data collaboratively and grant access to specific groups. To verify the integrity of this shared data, group members are required to sign each data block, ensuring that any unauthorized alterations can be detected. When a member is removed from the group, the remaining participants must re-sign the affected blocks to preserve both data integrity and security. However, most existing LBPA schemes [17–44] are tailored for individual DO's and do not accommodate auditing for shared data. As a result, they lack mechanisms for collaborative integrity verification and efficient user revocation. To address this limitation, there is a pressing need to develop advanced LBPA schemes that support secure and efficient auditing of shared data, along with dynamic user revocation capabilities in cloud storage systems.

5.4 | LBPA with Data Deduplication

Deduplication is a method to remove duplicate data, is widely applied in cloud storage to save bandwidth and storage space. For DOs to have a cost-effective storage option, CSPs must remove redundant data utilizing data deduplication. The existing LBPA schemes [17–44] support only data integrity. Hence, it is necessary to design LBPA schemes with data deduplication to ensure data integrity while eliminating duplicate data in cloud storage.

5.5 | LBPA with Blockchain Technology

In general, PA schemes assume that the TPA is trusted and honest and is responsible for verifying the integrity of the data stored in the cloud and ensuring that the CSP has not tampered with or misrepresented the data. However, if a malicious TPA colludes with a CSP to produce biased outcomes, the integrity of the entire scheme can be compromised. Such an attack is commonly

termed a collusion attack. To prevent potential collusion among malicious entities and enhance the trustworthiness of the auditing process, the LBPA schemes [23, 36] with blockchain technology have been designed. Blockchain can provide transparent and verifiable records of all auditing operations, thereby eliminating the need to fully trust the TPA or CSP. However, existing LBPA schemes integrated with blockchain still suffer from a single point of failure, as they are designed with a centralized TPA. Hence, it is necessary to design a decentralized blockchain with LBPA schemes for secure cloud storage. The blockchain's decentralized nature distributes control across multiple nodes, reducing the risk of collusion between the TPA and CSP. Each audit operation is recorded on the blockchain, ensuring transparency and verifiability for all parties involved. Therefore, by integrating blockchain technology with LBPA schemes, we can create a more secure, transparent, and reliable framework for data integrity verification in cloud storage environments.

5.6 | LBPA with Multi-Replica and Multi-Cloud Environments

Ensuring the availability and reliability of critical data often involves creating multiple copies and storing them on various cloud servers. This method protects against data loss from system failures, since the original file can be recovered from any of the intact replicas. However, the existing multi-replica scheme [41] stores all replicas on a single cloud server. If that server fails, all replicas become inaccessible. To address this issue, replicas should be distributed across different cloud servers, and their integrity should be audited simultaneously. Therefore, it is essential to develop multi-replica and multi-cloud LBPA schemes.

6 | Conclusion

This paper conducted a detailed survey of existing LBPA schemes, which are crucial for ensuring data integrity in cloud storage while resisting quantum attacks from quantum computers. In our survey, we provide a taxonomy by categorizing LBPA schemes into PKI-based, IBC-based, CLC-based, and ABC-based methods. Each technique is further classified based on hard problems such as SIS (ISIS) and R-LWE. Then, we provided the architecture, algorithms employed, and a comprehensive description of each LBPA scheme in the taxonomy. Furthermore, we analyzed the comparisons of these LBPA schemes based on security features and performance metrics, which are presented in tabular form. Finally, based on our survey and comparative analysis, we identified the challenges in the existing LBPA schemes. In our future work, we aim to design efficient and secure LBPA schemes to address these challenges and also enhance the security and reliability of cloud storage solutions.

6.1 | Future Research Directions

Future advancements in LBPA should focus on designing lightweight, trapdoor-free auditing mechanisms that maintain strong post-quantum security while reducing computational overhead. Incorporating module-lattice assumptions (e.g., M-LWE and M-SIS) represents a promising path toward

achieving higher efficiency and stronger resistance to algebraic attacks that affect classical R-LWE constructions.

Another important direction is the integration of blockchain technologies with LBPA to provide decentralized trust management, transparent auditability, and resilience against single-auditor failures. Additionally, adapting LBPA frameworks for multi-cloud, hybrid cloud, and cloud-edge environments is essential to support scalable, fault-tolerant, and distributed data storage architectures.

Finally, future research should incorporate simulation-based benchmarking, real-system prototyping, and experimental validation under realistic cloud settings to bridge the gap between theoretical security analysis and practical feasibility. To overcome these limitations, we propose module-LBPA schemes that rely on the M-LWE and M-SIS hardness assumptions.

6.2 | Real-World Deployment Considerations

Existing LBPA schemes can be readily integrated with major commercial cloud platforms such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud. In these schemes, the user begins by generating signatures for each file block and then uploads both the data blocks and their corresponding signatures to the real cloud platform. The CSP stores and manages this information using its standard infrastructure.

When data integrity needs to be verified, the TPA issues a random challenge to the CSP. The CSP responds by computing an integrity proof based on the stored data blocks and sending it back to the TPA. The TPA checks the correctness of the proof and reports the verification result to the user.

This workflow shows that LBPA schemes can be smoothly integrated into commercial cloud environments without altering existing storage interfaces or operational procedures. Consequently, the proposed LBPA schemes are practical for real-world use and can function efficiently within modern cloud ecosystems.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

References

1. J. R. Gudeme, S. K. Pasupuleti, and R. Kandukuri, "Review of remote data integrity auditing schemes in cloud computing: taxonomy, analysis, and open issues," *International Journal of Cloud Computing* 8, no. 1 (2019): 20–49.
2. C. T. Huang, L. Huang, Z. Qin, et al., "Survey on Securing Data Storage in the Cloud," *APSIPA Transactions on Signal and Information Processing* 3 (2014): e7.

3. Y. Zhu, H. Hu, G. J. Ahn, and M. Yu, "Cooperative provable data possession for integrity verification in multicloud storage," *IEEE Transactions on Parallel and Distributed Systems* 23, no. 12 (2012): 2231–2244.
4. C. C. Erway, A. Küpçü, C. Papamanthou, and R. Tamassia, "Dynamic provable data possession," *ACM Transactions on Information and System Security* 17, no. 4 (2015): 1–29.
5. C. Liu, J. Chen, L. T. Yang, et al., "Authorized public auditing of dynamic big data storage on cloud with efficient verifiable fine-grained updates," *IEEE Transactions on Parallel and Distributed Systems* 25, no. 9 (2013): 2234–2244.
6. C. Wang, S. S. Chow, Q. Wang, K. Ren, and W. Lou, "Privacy-preserving public auditing for secure cloud storage," *IEEE Transactions on Computers* 62, no. 2 (2011): 362–375.
7. J. Li, H. Yan, and Y. Zhang, "Identity-based privacy preserving remote data integrity checking for cloud storage," *IEEE Systems Journal* 15, no. 1 (2020): 577–585.
8. S. G. Worku, C. Xu, J. Zhao, and X. He, "Secure and efficient privacy-preserving public auditing scheme for cloud storage," *Computers and Electrical Engineering* 40, no. 5 (2014): 1703–1713.
9. S. More and S. Chaudhari, "Third party public auditing scheme for cloud storage," *Procedia Computer Science* 79 (2016): 69–76.
10. K. Yang and X. Jia, "An efficient and secure dynamic auditing protocol for data storage in cloud computing," *IEEE Transactions on Parallel and Distributed Systems* 24, no. 9 (2012): 1717–1726.
11. Q. Wang, C. Wang, K. Ren, W. Lou, and J. Li, "Enabling public auditability and data dynamics for storage security in cloud computing," *IEEE Transactions on Parallel and Distributed Systems* 22, no. 5 (2010): 847–859.
12. I. K. Meenakshi and S. George, "Cloud server storage security using TPA," *International Journal of Advanced Research in Computer Science & Technology (IJARCST)* (2014).
13. C. H. Ugwuishiwu, U. E. Orji, C. I. Ugwu, and C. N. Asogwa, "An overview of quantum cryptography and Shor's algorithm," *International Journal of Advanced Trends in Computer Science and Engineering* 9, no. 5 (2020): 8397–8405.
14. M. Kumar, "Post-quantum cryptography algorithms: Standardization and performance analysis," *Array* 15 (2022): 100242.
15. M. Ajtai, "Generating hard instances of lattice problems," in *Proceedings of the twenty-eighth annual ACM symposium on Theory of computing* (ACM, 1996), 99–108.
16. O. Regev, "On lattices, learning with errors, random linear codes, and cryptography," *Journal of the ACM* 56, no. 6 (2009): 1–40.
17. H. Liu and W. Cao, "Public proof of cloud storage from lattice assumption," *Chinese Journal of Electronics* 23, no. 1 (2014): 186–190.
18. X. Zhao, X. Wang, H. Xu, and Y. Wang, "Cloud data integrity checking protocol from lattice," *International Journal of High Performance Computing and Networking* 8, no. 2 (2015): 167–175.
19. X. Zhang, C. Xu, Y. Zhang, X. Zhang, and J. Wen, "Insecurity of a public proof of cloud storage from lattice assumption," *Chinese Journal of Electronics* 26 (2017): 88–92.
20. Y. Yan, L. Wu, G. Gao, H. Wang, and W. Xu, "A dynamic integrity verification scheme of cloud storage data based on lattice and Bloom filter," *Journal of Information Security and Applications* 39 (2018): 10–18.
21. W. Zhao, X. Jiang, and J. Wang, "Dynamic integrity verification of cloud storage data based on data proxy," in *Proceedings of the 2nd International Conference on Artificial Intelligence and Advanced Manufacture* (ACM, 2020), 517–522.
22. C. Zhou, L. Wang, and L. Wang, "Lattice-based provable data possession in the standard model for cloud-based smart grid data management systems," *International Journal of Distributed Sensor Networks* 18, no. 4 (2022): 15501329221092940.
23. Z. Li, T. Zhang, D. Zhao, Y. Zha, and J. Sun, "Post-Quantum Privacy-Preserving Provable Data Possession Scheme Based on Smart Contracts," *Wireless Communications and Mobile Computing* 2023, no. 1 (2023): 8772700.
24. Y. Yang, Q. Huang, and F. Chen, "Secure cloud storage based on RLWE problem," *IEEE Access* 7 (2018): 27604–27614.
25. X. Zhang and C. Xu, "Efficient identity-based public auditing scheme for cloud storage from lattice assumption," in *2014 IEEE 17th International Conference on Computational Science and Engineering IEEE (IEEE, 2014)*, 1819–1826.
26. X. Zhang, C. Xu, and C. Jin, "Enabling identity-based cloud storage public auditing with quantum computers resistance," *International Journal of Electronic Security and Digital Forensics* 8, no. 1 (2016): 82–98.
27. Z. Liu, Y. Liao, X. Yang, Y. He, and K. Zhao, "Identity-based remote data integrity checking of cloud storage from lattices," in *2017 3rd International Conference on Big Data Computing and Communications (BIG-COM)* (IEEE, 2017), 128–135.
28. X. Zhang, J. Zhao, C. Xu, H. Wang, and Y. Zhang, "DOPIV: Post-quantum secure identity-based data outsourcing with public integrity verification in cloud storage," *IEEE Transactions on Services Computing* 15, no. 1 (2019): 334–345.
29. X. Zhang, H. Wang, and C. Xu, "Identity-based key-exposure resilient cloud storage public auditing scheme from lattices," *Information Sciences* 472 (2019): 223–234.
30. H. Li, L. Liu, C. Lan, C. Wang, and H. Guo, "Lattice-based privacy-preserving and forward-secure cloud storage public auditing scheme," *IEEE Access* 8 (2020): 86797–86809.
31. X. Zhang, C. Huang, Y. Zhang, J. Zhang, and J. Gong, "Ldvas: Lattice-based designated verifier auditing scheme for electronic medical data in cloud-assisted wbans," *IEEE Access* 8 (2020): 54402–54414.
32. H. Wang, X. A. Wang, J. Liu, and C. Lin, "Identify-Based Outsourcing Data Auditing Scheme with Lattice," in *Frontiers in Cyber Security: Third International Conference, FCS 2020*, vol. 2020 (Springer Singapore, 2020), 347–358.
33. X. Liu, Y. Luo, X. Yang, L. Wang, and X. Zhang, "Lattice-based proxy-oriented public auditing scheme for electronic health record in cloud-assisted wbans," *IEEE Systems Journal* 16, no. 2 (2022): 2968–2978.
34. Q. Wang, C. Cheng, R. Xu, J. Ding, and Z. Liu, "Analysis and Enhancement of a Lattice-Based Data Outsourcing Scheme with Public Integrity Verification," *IEEE Transactions on Services Computing* 15, no. 4 (2022): 2226–2231.
35. S. Yang and J. Chang, "Identity-based remote data integrity auditing from lattices for secure cloud storage," *Cluster Computing* 27 (2024): 1–12.
36. D. Gautam, S. Prajapat, P. Kumar, A. K. Das, K. Cengiz, and W. Susilo, "Blockchain-assisted post-quantum privacy-preserving public auditing scheme to secure multimedia data in cloud storage," *Cluster Computing* 27 (2024): 1–14.
37. Y. Yang, Y. Sun, Q. Huang, W. Yin, and F. Chen, "RLWE-Based ID-DIA protocols for cloud storage," *IEEE Access* 7 (2019): 55732–55743.
38. S. M. Basha, C. S. Bindu, C. Sasikala, and P. D. K. Reddy, "Certificateless Public Auditing Protocol in Cloud," *International Journal of Research in Advent Technology* 6, no. 5 (2018): 1–5.
39. C. Sasikala and C. S. Bindu, "Certificateless batch verification protocol to ensure data integrity in multi-cloud using lattices," *International Journal of Internet Protocol Technology* 12, no. 4 (2019): 236–242.
40. C. Sasikala and C. Shoba Bindu, "Certificateless remote data integrity checking using lattices in cloud storage," *Neural Computing and Applications* 31 (2019): 1513–1519.
41. Y. Zhang, Y. Sang, Z. Xi, and H. Zhong, "Lattice based multi-replica remote data integrity checking for data storage on cloud," in *Parallel*

Architectures, Algorithms and Programming: 10th International Symposium, PAAP 2019, Guangzhou, China, December 12–14, 2019, Revised Selected Papers 10 (Springer, 2020), 440–451.

42. H. Li, Y. Wang, X. Fu, et al., “PSCPAC: Post-quantum secure certificateless public auditing scheme in cloud storage,” *Journal of Information Security and Applications* 61 (2021): 102927.

43. R. Cheeturi, S. K. Pasupuleti, and R. R. Rout, “Lattice-Based Efficient Certificateless Public Auditing Scheme from RLWE for Cloud Storage,” in *2024 IEEE 24th International Symposium on Cluster, Cloud and Internet Computing Workshops (CCGridW)* (IEEE, 2024), 26–33.

44. X. Zhang, X. Liu, Q. Liu, and J. Wang, “Revocable Attribute-Based Data Integrity Auditing Scheme on Lattices,” in *2022 International Conference on Computer Science, Information Engineering and Digital Economy (CSIEDE 2022)* (Atlantis Press, 2022), 383–396.

45. H. Minkowski, *Geometrie der Zahlen* (Teubner, 1910).

46. D. Micciancio and S. Goldwasser, *The Complexity of Lattice Problems: A Cryptographic Perspective* (Springer, 2002).

47. V. Lyubashevsky, C. Peikert, and O. Regev, “On ideal lattices and learning with errors over rings,” in *Annual International Conference on the Theory and Applications of Cryptographic Techniques (EUROCRYPT)* (Springer, 2010), 1–23.

48. R. Lindner and C. Peikert, “Better key sizes (and attacks) for LWE-based encryption,” in *Cryptographic Hardware and Embedded Systems (CHES)* (Springer, 2011), 319–339.

49. D. Micciancio and O. Regev, “Worst-case to average-case reductions based on Gaussian measures,” *SIAM Journal on Computing* 37, no. 1 (2007): 267–302.

50. C. Gentry, C. Peikert, and V. Vaikuntanathan, “Trapdoors for hard lattices and new cryptographic constructions,” in *STOC* (Foundational trapdoor construction (GPV), 2008).

51. D. Micciancio and C. Peikert, “Trapdoors for Lattices: Simpler, Tighter, Faster, Smaller,” in *EUROCRYPT* (Gadget trapdoors and practical trapdoor constructions, 2012).

52. V. Lyubashevsky, “Lattice Signatures Without Trapdoors,” in *Advances in Cryptology – EUROCRYPT 2012*, vol. 7237 Lecture Notes in Computer Science (Springer, 2012), 738–755.

53. C. Peikert, “A Decade of Lattice Cryptography,” *Foundations and Trends in Theoretical Computer Science* 10, no. 4 (2016): 283–424. Survey; foundational reference.

54. S. S. Al-Riyami and K. G. Paterson, “Certificateless public key cryptography,” in *International Conference on the Theory and Application of Cryptology and Information Security* (Springer, 2003), 452–473.

55. H. K. Maji, M. Prabhakaran, and M. Rosulek, “Attribute-based signatures,” in *Cryptographers’ track at the RSA conference* (Springer, 2011), 376–392.

56. A. Mukherjee, A. Aikata, A. C. Mert, et al., “ModHE: Modular Homomorphic Encryption Using Module Lattices: Potentials and Limitations,” 2023, Cryptology ePrint Archive, Paper, <https://eprint.iacr.org/2023/895>.

57. S. Liu and A. Sakzad, “CRYSTALS-Kyber With Lattice Quantizer,” 2024, arXiv preprint arXiv:2401.15534.

58. K. de Boer and W. van Woerden, “Lattice-based Cryptography: A survey on the security of the lattice-based NIST finalists,” 2025, Cryptology ePrint Archive, Paper 2025/304.

59. M. E. Sabani, I. K. Savvas, and G. Garani, “Learning with Errors: A Lattice-Based Keystone of Post-Quantum Cryptography,” *Signals* 5, no. 2 (2024): 216–243.

Appendix A

Algorithms

PKI-based LBPA Schemes

PKI-based schemes consist of five algorithms.

- *KeyGen* ($1^n \rightarrow pk, sk$): The DO takes the input security parameter 1^n , runs the TrapGen algorithm, and outputs a public-secret key pair (pk, sk) .
- *SignGen* ($sk, F \rightarrow \Phi$): And the DO also takes the input with sk and the data file F , runs the SamplePre algorithm, and outputs a group of signatures Φ .
- *ChallengeGen* ($F_{id} \rightarrow C$): The TPA runs this algorithm; it takes input F and file identity F_{id} (e.g., file name, total number of blocks, and so on) and outputs a challenge message C .
- *ProofGen* ($F, \Phi, C \rightarrow P$): The CSP takes the input file F , the set of signatures Φ , and the challenge message, C runs the SamplePre algorithm, and outputs a proof P .
- *ProofVerify* ($pk, C, P \rightarrow \text{true/false}$): TPA executes this algorithm by taking the input pk , challenge C , and proof P and outputs the result as either true or false.

IBC-based LBPA Schemes

The IBC-based schemes consist of six algorithms: Setup, KeyGen, SignGen, ChallengeGen, ProofGen, and ProofVerify.

- *Setup* ($1^n \rightarrow msk, mpk, pp$): The KGC takes the security parameter 1^n as input, runs the TrapGen algorithm, and outputs the master secret key (msk), the master public key (mpk), and the system public parameters (pp).
- *KeyGen* ($ID, msk, pp \rightarrow sk$): The DO inputs their identity ID , along with msk and pp , and runs the NewBasisDel algorithm to generate a sk for the DO.
- *SignGen*, *ChallengeGen*, *ProofGen*, and *ProofVerify*: These algorithms run in the same way as in the PKI-based LBPA schemes.

CLC-based LBPA Schemes

The CLC-based schemes contain nine algorithms: Setup, ExtractpartialsecretKey, SetsecretValue, SetsecretKey, SetPublicKey, SignGen, ChallengeGen, ProofGen, and ProofVerify.

- *Setup* ($1^n \rightarrow msk, mpk, pp$): The KGC takes the input security parameter 1^n , runs the TrapGen algorithm, and outputs the master secret-public key pair (msk, mpk) and pp .
- *Extractpartialsecretkey* ($pp, msk, ID \rightarrow P_{ID}$): The KGC also takes inputs, pp , the msk , DO’s identity ID , runs the NewBasisDel algorithm to generate the partial secret key P_{ID} .
- *SetsecretValue* ($pp, ID \rightarrow S_{ID}$): This algorithm is run by the DO; it takes pp the DO’s identity, ID , selects a random matrix, and produces a random short lattice basis as his secret value S_{ID} .
- *Setsecretkey* ($pp, P_{ID}, S_{ID} \rightarrow sk$): It is run by the DO. DO takes inputs pp , the partial secret key P_{ID} , and the secret value S_{ID} , then it combines the P_{ID} with S_{ID} and sets its full sk .
- *SetPublickey* ($pp, S_{ID} \rightarrow pk$): It is run by the DO. It takes the pp and S_{ID} as input and outputs pk . The *SignGen*, *ChallengeGen*, *ProofGen*, and *ProofVerify* algorithms work the same as the ABC-based LBPA schemes.

ABC-based LBPA Schemes

It consists of six algorithms like IBC-based LBPA, such as Setup, KeyGen, SignGen, ChallengeGen, ProofGen, and ProofVerify.

- *Setup* ($1^n \rightarrow msk, mpk, pp$): It works like the IBC-based LBPA.
- *KeyGen* ($pp, ID, msk, \mathcal{A} \rightarrow sk$): The $\mathcal{TA}$ takes pp , DO identity ID , attribute set $\mathcal{A}$, msk , runs the NewBasisDel algorithm, and outputs sk .
- *SignGen* ($sk, y, F \rightarrow \mathcal{T}, \phi$): The DO takes inputs sk , access policy y , data file F , runs the SamplePre algorithm, and outputs the file tag $\mathcal{T}$, and signature set ϕ .
*Access policy: The access policy specifies the set of properties that a signer requires in order to properly establish a valid signature.
- *ChallengeGen* ($mpk, \beta, \mathcal{T} \rightarrow C$): The TPA runs this algorithm by taking the mpk , the TPA 's attribute set B , the file tag $\mathcal{T}$, and then generate the challenge C .
- *ProofGen* ($mpk, F, \mathcal{T}, \Phi, C \rightarrow \mathcal{P}$): The CSP runs this algorithm; it takes inputs mpk , a data file F , file tag $\mathcal{T}$, signature set Φ , and outputs the proof $\mathcal{P}$.
- *ProofVerify* ($mpk, \beta, C, \mathcal{P} \rightarrow true/false$): It takes the mpk , TPA 's attribute set β , challenge C , proof $\mathcal{P}$, and outputs the audit result as true or false.